

A fuzzy group decision-making method integrating cooperative game theory and quantum cognition for dynamic trust network

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ABSTRACT

In group decision-making within social networks, the process often exhibits high complexity and uncertainty due to psychological cognitive differences, fluctuations in trust relationships, and mutual interference effects among decision-makers (DMs). To address this issue, this paper proposes a fuzzy group decision-making method that integrates cooperative game theory with quantum cognition. Firstly, an attribute weighting model is constructed by integrating psychological preference perception with individual contribution, so as to determine the attributes' weights of DMs under different psychological states. Secondly, a quantum intuitionistic fuzzy set is proposed to represent the dynamic fuzziness of trust relationships, and the interactive evolution mechanism of trust and contribution is characterized through a two-stage trust correction model. Furthermore, Kullback-Leibler divergence is used to measure the interference effect between DMs, and the quantum probabilities of alternatives are computed to derive the final ranking of alternatives. Finally, the effectiveness of this method was verified by taking the emergency response to meteorological disasters as a case. The results show that the psychological preferences of DMs and marginal contributions present a significant dynamic coupling effect in the interaction process. The changes in psychological preference perception will further affect the evolution of group contribution distribution and trust mechanisms. This result verifies the effectiveness of the model in depicting psychological differences and trust dynamics, and can achieve more reasonable group ranking in complex social network.

1. Introduction

With the increase in information complexity and the diversity of stakeholders, the importance of multi-attribute group decision-making (MAGDM) in solving complex problems has become increasingly prominent (Liu et al., 2025). Especially in emergency situations such as meteorological disasters, DMs need to make trade-offs by integrating multiple conflicting attributes within a limited time (Gayen et al., 2024; Meng et al., 2024; Xu et al., 2025b). For instance, in the process of disaster emergency response, factors such as the accuracy of weather forecasts, the extent of the disaster, the distribution of resources, and the priority demands of different departments and groups need to be taken into account simultaneously. During this process, the high uncertainty of the decision-making environment and the interactivity among DMs make MAGDM a necessary scientific decision-making tool (Fallah et al., 2023; Jiang et al., 2024; Zhou et al., 2025).

In MAGDM, the reasonable allocation of attribute weights directly determines the scientificity and stability of the decision results

(Teng et al., 2024; Wu et al., 2024). In a complex group decision-making environment, the cooperative contribution of DMs (Meng et al., 2023b; Zhou et al., 2024) and psychological states (such as risk perception and trust attitude) often interact and evolve dynamically: Psychological preferences (Chao et al., 2021; Van Dijk & De Dreu, 2021; Yang et al., 2022) can change their participation degree and judgment weight, while the cooperative relationship in the group will in turn regulate the individual's cognition and trust level. Therefore, a coupling mechanism that encompasses both is crucial for truly reflecting the decision-making behavior of a group. However, traditional methods represented by AHP and entropy weight method (Han et al., 2023; Khosravi & Izbi-rak, 2024; Xin & Ying, 2024) assume that the evaluation results of each DM can be directly weighted and summarized. This static framework lacks state feedback and interaction terms, and cannot describe the non-linear impact of individual judgment on group consensus when risk perception, trust level or decision-making pressure changes. Therefore, it is difficult to depict the weight transfer effect caused by psychological fluctuations. Although the weighting methods based on intelligent

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optimization (such as PSO optimization (Rani & Garg, 2023)) introduce adaptive search mechanisms, their fitness functions only minimize global errors or maximize information consistency, making it difficult to embed the contribution differences or trust coupling among DMs. Therefore, the weights can only be adjusted at the numerical level. However, it is unable to express the dynamic synergy effect in group interaction. Consequently, existing methods suffer from structural limitations in capturing the intertwined “psychological-cooperative” dynamics, reducing their explanatory power in heterogeneous groups and trust-evolving environments.

In the context of modern social networks, trust relationships play a crucial role in MAGDM, directly influencing the dissemination and sharing of information as well as the formation and execution of decisions (Cao et al., 2025; Ji et al., 2024; Sun et al., 2023b). The trust relationship in reality has the characteristics of fuzziness, time-variability and bidirectionality, and its state often fluctuates dynamically between trust and distrust (Liu et al., 2023a; Xing et al., 2023; Zhu et al., 2025). However, existing models typically view trust relationships as fixed parameters or one-way weights, which are used to adjust the importance of information or the weights of DMs. This static setting lacks a trust feedback mechanism in structure and is unable to describe the trust correction effect that individuals experience during the information interaction process due to differences in opinions, changes in cooperation effectiveness or risk expectations (Ma et al., 2024). Furthermore, although some studies based on network or graph structures have taken trust propagation into account, they mostly adopt linear superposition or threshold update rules (Guo et al., 2024; Wu et al., 2023b), ignoring the possible superposition and interference effects of trust relationships under multi-party interactions. As a result, most existing trust models can only capture averaging or trend-level changes, but fail to reflect the nonlinear evolution and dynamic coupling characteristics of DMs’ trust in reality.

Quantum probability theory (QPT) provides a new theoretical tool for solving complex decision-making behavior problems (Busemeyer, 2012; Wang et al., 2024). The core idea is to describe cognitive non-independence and psychological superposition effects through probability amplitude and phase interference (Xu et al., 2025a; Yan et al., 2025), that is, in the decision-making process, the mutual influences among different options will change the DM’s evaluation of these options (Zhu et al., 2026). Unlike the classical probability assumption that events are independent, quantum models can reflect the interference relationships among different decision-making paths, revealing the nonlinear impacts of psychological state changes, trust evolution, and cognitive conflicts on decision-making outcomes (Gao et al., 2025; Liu & Zhu, 2022; Xiao, 2022). On this basis, the quantum-like Bayesian network (QLBN) was proposed to handle the mutual interference among DMs (Wu et al., 2022; Yan et al., 2024). However, the existing QLBN framework still has limitations in the measurement of interference effects: Most studies use cosine similarity (Jiang & Liu, 2022; Liu et al., 2023b) or entropy (Han & Liu, 2023; She et al., 2021; Xiao et al., 2025) methods as approximate indicators of the interference angle. The former only reflects the directional similarity of the preference vector and ignores the differences in the score distribution, while the latter focuses on the dispersion degree of individual evaluations and cannot express the information asymmetry and nonlinear differences among the preference distributions of different DMs.

In summary, while existing studies have addressed the application scope and effectiveness of MAGDM, several key issues remain unresolved: 1) Traditional weight determination methods struggle to simultaneously account for both the psychological states and cooperative relationships of DMs, resulting in an inability to fully capture the true intentions and interactions of DMs in complex decision environments. 2) In modern social networks, the trust relationship between DMs is often fuzzy and dynamic, and the traditional decision model fails to fully capture the dynamic uncertainty, thus affecting the quality of decision making. 3) Although quantum probability theory demonstrates unique advantages in handling complex decision behaviors, particularly inter-

ference effects, current measurement methods—such as cosine similarity and entropy—are insufficient for accurately quantifying the complex interference effects between DMs, which undermines the model’s precision.

Therefore, it is necessary to integrate three theoretical mechanisms under a unified framework: 1) Quantum intuitionistic fuzzy sets (QIFS) provide state superposition and probability amplitude expression for trust relationships, enabling the fuzzy states of trust, distrust, and hesitation to dynamically evolve in quantum form; 2) Based on the principle of quantum probability theory, a measurement system that can simultaneously reflect the intensity and directionality of interference is constructed to quantify the cognitive interference and non-independent effects among DMs. 3) Cooperative game theory takes marginal contribution as its core, characterizes the relative importance of individuals in collaborative decision-making, and provides a game-like basis for quantifying psychological preferences and correcting trust. On this basis, this paper proposes a multi-attribute quantum group decision making model based on cooperative game theory. The primary contributions are as follows:

1) A psychologically interpretable cooperative weighting mechanism is proposed by embedding a Sigmoid-type preference function into the Shapley-based framework, enabling the dynamic linkage between DMs’ psychological states and their cooperative contributions.

2) A quantum intuitionistic fuzzy representation of trust relationships is constructed, extending traditional IFS into a quantum state form that captures the coexistence and evolution of trust and distrust under uncertainty.

3) A phase mapping function is constructed based on the Kullback-Leibler divergence to measure the interference Angle between preference vectors, thereby revealing the cognitive interference effect and improving the interpretability of group preference fusion in the quantum probability space.

The overall structure of this paper is arranged as follows. Section 2 provides the theoretical preliminaries, including Intuitionistic fuzzy set, social networks, and quantum probability theory, which together constitute the basis for the subsequent model construction. Section 3 describes the flow structure of the paper. In Section 4, a quantum group decision model based on cooperative game is proposed. Section 5 validates the model through a case analysis of a Forest fire and demonstrates its robustness through sensitivity analysis and comparative analysis. Section 6 discusses the research results and, in combination with the model conclusions, puts forward policy suggestions with practical guiding significance. Finally, Section 7 presents the conclusions of the paper and outlines prospects for future research.

2. Preliminaries

2.1. Intuitionistic fuzzy set

The concept of intuitionistic fuzzy set (IFS) was first proposed by Atanassov (1986). IFS extends the classical fuzzy set by introducing an additional dimension of hesitation to explicitly represent uncertainty and incompleteness in information.

Formally, an intuitionistic fuzzy number is expressed as a triplet $Q = \langle \mu_Q(x), \nu_Q(x), \pi_Q(x) \rangle$, where $\mu_Q(x)$ denotes the membership degree, representing the extent to which element x belongs to the set Q ; $\nu_Q(x)$ denotes the non-membership degree, reflecting the extent to which x does not belong to Q ; and $\pi_Q(x) = 1 - \mu_Q(x) - \nu_Q(x)$ quantifies the hesitancy degree, which measures the indeterminacy or cognitive ambiguity regarding the inclusion of x in Q .

In this paper, the IFS is adopted as the fundamental representation for modeling trust relationships among DMs (Sun et al., 2023a). Specifically, the membership degree $\mu_Q(x)$ is used to describe the level of trust, the non-membership degree $\nu_Q(x)$ expresses the level of distrust, and the hesitancy degree $\pi_Q(x)$ characterizes the indeterminate or ambiguous state between trust and distrust.

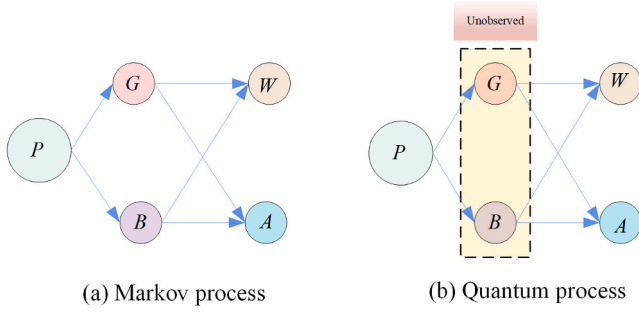


Fig. 1. Markov chain path diagram.

2.2. Social network (SN)

Definition 1 (Wang et al., 2024). A social network can be described as a directed graph $G(D, E)$, in which the node set $D = \{d_1, d_2, \dots, d_s\} (s \geq 3)$ corresponds to the group of DMs, and the edge set $E = \{e_1, e_2, \dots, e_q\}$ contains all directed links among them. If an ordered pair $(e_k, e_h) \in E$ exists, it indicates that DM d_k exerts a direct influence on DM d_h ($k, h \in s$). The network structure can be encoded by a sociometric matrix $T = (t_{kh})_{s \times s}$, where each entry satisfies $t_{kh} \in \{0, 1\}$ and is defined as follows:

$$t_{kh} = \begin{cases} 1, & (d_k, d_h) \in E \\ 0, & \text{other} \end{cases} \quad (1)$$

$k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h.$

2.3. Quantum probability theory

Quantum probability theory is a probability calculation framework distinct from classical probability theory. It originated from quantum mechanics theory and is widely applied to handle decision-making problems with significant complexity, uncertainty, and mutual interference (Busemeyer, 2012). Unlike classical probability theory, which relies on deterministic sample Spaces and probability axiom systems, quantum probability theory is based on Hilbert space mathematically and describes the uncertainty characteristics of events through quantum states and measurement operators. It can characterize non-classical probability phenomena caused by cognitive interference, information superposition, or decision conflicts, Fig. 1 visually demonstrates the difference where the classical model assumes that paths are independent, while the quantum model allows paths to interfere with each other.

In quantum probability theory, suppose that the state space of a decision problem is a complex Hilbert space $\mathcal{H}$, then any quantum state can be represented as a normalized vector (Zhu et al., 2026). For a finite-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^n$, where $\mathbb{C}$ denotes the complex field and $\mathbb{C}^n$ represents an n -dimensional complex vector space composed of n complex components, the quantum state of the system $|\psi\rangle$ can be expressed as the following column vector:

$$|\psi\rangle = \sum_{i=1}^n \alpha_i |e_i\rangle \quad (2)$$

where, $|e_i\rangle$ denotes the standard orthonormal basis vector in the Hilbert space, and $\alpha_i \in \mathbb{C}$ is the complex amplitude, satisfying the normalization condition: $\sum_{i=1}^n |\alpha_i|^2 = 1$, and $|\alpha_i|^2$ represents the probability of measuring the state $|\psi\rangle$ and obtaining the outcome corresponding to $|e_i\rangle$.

3. Problem description

The primary objective of a MAGDM problem is to obtain a consensus-oriented solution from a set of alternatives evaluated under multiple attributes. The essential components of a MAGDM problem can

be summarized as follows: 1) DMs. Let $D = \{d_1, d_2, \dots, d_s\} (s \geq 3)$ denote the set of DMs. Their associated weights are represented as $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$, where the conditions $\lambda_k \geq 0 (k = 1, 2, \dots, s)$ and $\sum_{k=1}^s \lambda_k = 1$ hold. 2) Alternatives. The set of possible solutions is given by $A = \{a_1, a_2, \dots, a_m\} (m \geq 2)$, where each a_i represents a candidate option to be evaluated. 3) Attributes. The attribute for evaluation are defined as $C = \{c_1, c_2, \dots, c_n\} (n \geq 2)$. For each DM d_k , the attributes' weights set is expressed as $W^k = \{\omega_1^k, \omega_2^k, \dots, \omega_n^k\}$, subject to the constraints $\omega_j^k \geq 0 (j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ and $\sum_{j=1}^n \omega_j^k = 1$.

For ease of reference, the main mathematical symbols used in this paper are summarized in Table 1.

3.1. Determination of decision information

1) For each DM $d_k (k = 1, 2, \dots, s)$, the assessment of alternative $a_i (i = 1, 2, \dots, m)$ with respect to attribute $c_j (j = 1, 2, \dots, n)$ is denoted by a crisp number $\hat{z}_{ij}^k$. The collection of all evaluations provided by DM $d_k (k = 1, 2, \dots, s)$ forms an evaluation information matrix, expressed as $\hat{Z}^{(k)} = (\hat{z}_{ij}^k)_{m \times n}$, ($k = 1, 2, \dots, s$), as follows:

$$\hat{Z}^{(k)} = \begin{pmatrix} \hat{z}_{11}^k & \hat{z}_{12}^k & \dots & \hat{z}_{1n}^k \\ \hat{z}_{21}^k & \hat{z}_{22}^k & \dots & \hat{z}_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ \hat{z}_{m1}^k & \hat{z}_{m2}^k & \dots & \hat{z}_{mn}^k \end{pmatrix}, k = 1, \dots, s. \quad (3)$$

2) In MAGDM, DMs may adopt heterogeneous evaluation standards across different attributes. To eliminate such inconsistencies, normalization is applied so as to reduce scale effects, avoid undue dominance of any single DM's assessment, and guarantee fairness in the aggregation process. The normalized evaluation matrix corresponding to DM d_k is denoted as $Z^{(k)} = (z_{ij}^k)_{m \times n}$, ($k = 1, 2, \dots, s$), and can be expressed as follows:

$$z_{ij}^k = \begin{cases} \frac{\hat{z}_{ij}^k - \min_i \hat{z}_{ij}^k}{\max_i \hat{z}_{ij}^k - \min_i \hat{z}_{ij}^k}, & c_j \text{ is a benefit attribute;} \\ \frac{\max_i \hat{z}_{ij}^k - \hat{z}_{ij}^k}{\max_i \hat{z}_{ij}^k - \min_i \hat{z}_{ij}^k}, & c_j \text{ is a cost attribute;} \end{cases} \quad (4)$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s.$

3.2. The quantum decision-making framework

This paper constructs a multi-attribute quantum group decision-making framework to comprehensively characterize the decision-maker's evaluation information, psychological preference regulation, trust relationship evolution, and quantum reasoning process under a unified modeling structure, thereby reflecting the interaction of multiple cognitive factors in group decision-making. The overall process of this framework is shown in Fig. 2, mainly including the following four interrelated steps:

1) Collect the initial evaluation information of each DM, and normalize the original evaluation matrix according to a unified standard to provide a consistent data basis for subsequent multi-attribute evaluation and weight calculation.

2) Based on the normalized evaluation information, the marginal contribution of decision-makers under different alliance structures is characterized by the Shapley value in cooperative game. Meanwhile, in order to reflect the influence of an individual's psychological state on the degree of attribute attention during the decision-making process, a psychological preference function is constructed to dynamically adjust the attributes' weights. On the basis, an attributes' weights optimization model is established, thereby obtaining the optimal weight distribution jointly constrained by psychological preferences and cooperative contributions.

Table 1
Nomenclature table.

Symbol	Description	Symbol	Description
$D = \{d_1, d_2, \dots, d_s\}$	Set of DMs ($s \geq 3$)	$\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_s\}$	Set of DMs' weights.
$A = \{a_1, a_2, \dots, a_m\}$	Set of alternatives ($m \geq 2$)	$C = \{c_1, c_2, \dots, c_n\}$	Set of attributes ($n \geq 2$).
$W^k = \{\omega_1^k, \dots, \omega_n^k\}$	Set of attributes' weights.	$Z^{(k)} = (z_{ij}^k)_{m \times n}$	Initial evaluation matrix of d_k .
$Z^{(k)} = (z_{ij}^k)_{m \times n}$	Normalized evaluation matrix of d_k .	$\sigma_j^k(\xi_k)$	Attribute' weight considering psychological preference.
ξ_k	Psychological preference factor of d_k .	χ	Sensitivity parameter controlling Sigmoid steepness.
ξ_0	Critical point of the psychological preference factor.	ϕ_k	Shapley value of DM d_k .
CTL_k	Comprehensive trust level at Stage 1.	$\alpha_{kh}, \beta_{kh}, \gamma_{kh}$	Parameters of QIFS.
CTL_k	Comprehensive trust level at Stage 2.	ε	Permissible deviation in fairness constraint.
α	Adjustment factor ($\alpha \in [0, 1]$).	β	Influence coefficient ($\beta \in [0, 1]$).
η	Scale factor linking trust and weight.	δ_k	Preference vector of d_k .
$D_{KL}(d_k \ d_h)$	KL divergence between d_k and d_h .	$D'_{KL}(d_k \ d_h)$	Symmetric KL divergence between d_k and d_h .
θ_{kh}	Interference angle between d_k and d_h .	$p(d_k)$	Prior probability in quantum-like Bayesian network.
$p(a_i d_k)$	Conditional probability of a_i given d_k .	$p(a_i)$	Quantum probability of alternative a_i .

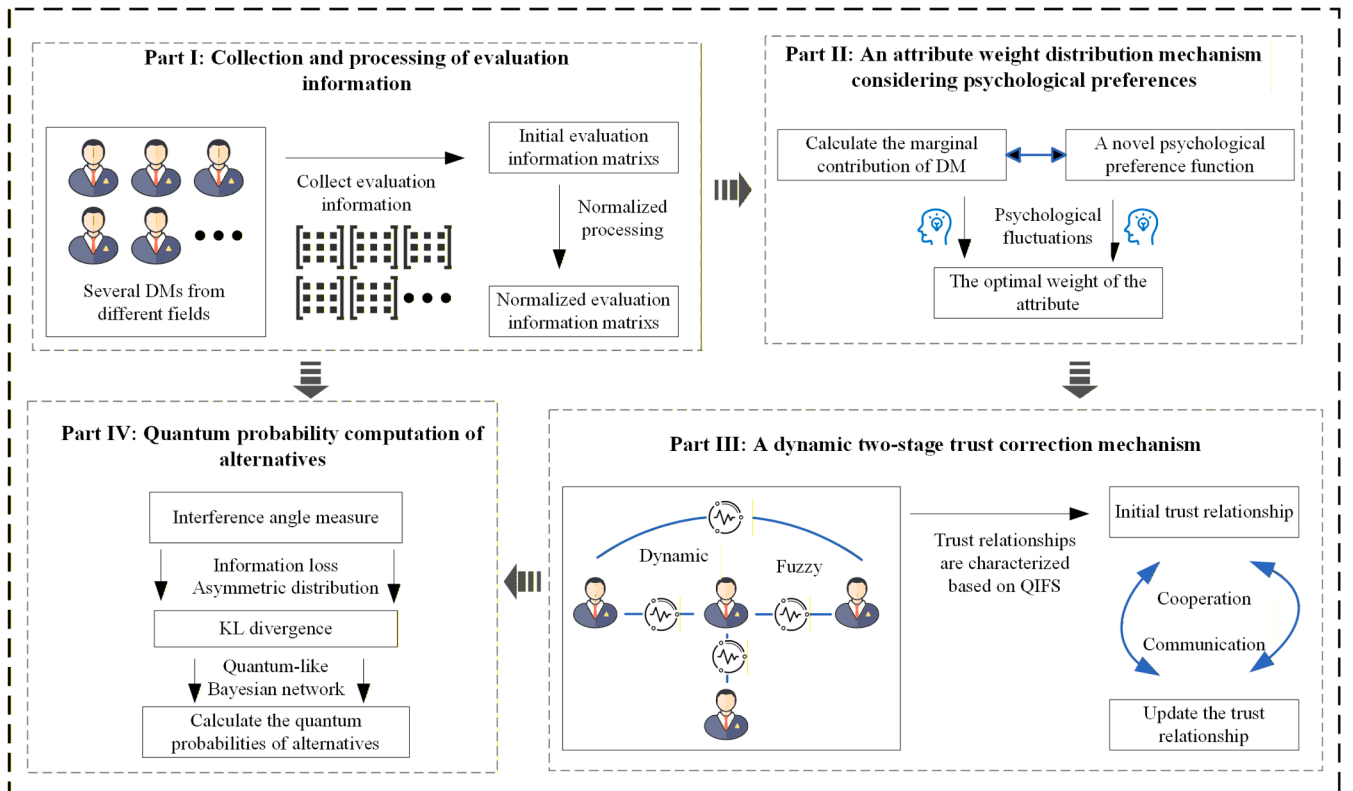


Fig. 2. The resolution framework of this paper.

3) According to the dynamic and fuzzy characteristics of the trust relationship between decision makers, a quantum intuitionistic fuzzy set is proposed to represent the trust relationship, and a two-stage dynamic trust correction mechanism is constructed, so that the trust level can be updated according to the marginal contribution of DMs. Through this process, the DM's weights that reflects the comprehensive influence of the DM is further determined.

4) On this basis, a quantum-like Bayesian network is constructed, and the multi-attribute evaluation information adjusted by the DM's weight and psychological preference is respectively used as the basis for generating prior probability and conditional probability. Meanwhile, the symmetrical KL divergence is utilized to characterize the differences between the preference distributions of DMs and map them to interference angles, thereby calculating the quantum probabilities of each alternative scheme and completing the ranking.

4. A quantum group decision-making method considering psychological preference and cooperative relationship

4.1. A weight allocation model for attributes considering psychological preference

This section aims to study the influence mechanism of decision-makers' psychological preferences on the behavior of attribute weight allocation. In a complex and uncertain group decision-making environment, the distribution of decision-makers' attention and the degree of attribute emphasis are often influenced by factors such as psychological state, cognitive stress and risk perception, and show obvious non-linear adjustment characteristics. To depict this phenomenon, this paper first constructs a psychological preference function based on the Sigmoid structure to describe the process by which the changes in the psychological state of decision-makers affect the adjustment of attribute weights.

On this basis, the attribute weights adjusted by psychological preferences are further introduced into the cooperative game and weight optimization framework, thereby achieving an adaptive response of weight distribution to individual behavioral preferences.

4.1.1. A novel psychological preference function

From the perspective of behavioral decision-making theory, in complex and uncertain environments, the psychological preferences of DMs usually do not change linearly but undergo an evolutionary process from neglect to reinforcement and then to stabilization. This process reflects the typical behavioral characteristics of DMs, who are slow to respond to attribute information in the early stage of cognition, highly sensitive to key information in the middle stage, and tend to be saturated after their preferences gradually stabilize. The S-shaped structure of the Sigmoid function (Dombi & Jónás, 2022) can well describe the above-mentioned nonlinear cognitive adjustment process, which is mainly reflected in the following aspects: (1) When the preference level is low, the response to external stimuli is relatively limited; (2) Within the middle range, preferences rapidly amplify with changes in psychological states, demonstrating significant sensitivity. (3) Gradually converge after the preference stabilizes to avoid the unbounded growth of weights.

In addition, the Sigmoid function has good smoothness and boundedness, which can ensure the continuity and stability of the psychological preference adjustment process. This is consistent with the behavioral characteristics of DMs who gradually adjust the allocation of cognition and attention under conditions of risk, pressure and incomplete information. In contrast, linear functions are difficult to reflect the non-uniform changes in psychological preferences within sensitive intervals, while exponential functions or piecewise functions often lack symmetry or continuous differentiability, which is not conducive to depicting a smooth psychological adjustment process.

On the basis, this paper constructs a psychological preference function based on Sigmoid's S-shaped adjustment mechanism to describe the influence of DMs' psychological states on the allocation of attribute weights. Specifically, the psychological state variable ξ_k is introduced to represent the psychological preference level of the DM d_k , and it is mapped to the attribute weights under the effect of psychological preferences through the Sigmoid structure, thereby achieving a smooth transition of weights from the rational state to the psychological regulatory state. The psychological preference function is defined as follows:

$$\omega_j^k(\xi_k) = \frac{\omega_j^k}{1 + e^{-\chi(\xi_k - \xi_0)}}, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (5)$$

where, $\omega_j^k(j = 1, 2, \dots, n, k = 1, 2, \dots, s)$ is the weight of the attribute of a known DM's rational state, and $\xi_k(k = 1, 2, \dots, s)$ is the psychological preference factor of DM $d_k(k = 1, 2, \dots, s)$, reflecting the DM's mental state, preference or perception level. $\chi(\chi \geq 0)$ is the sensitivity parameter that controls the steepness of the Sigmoid function. The larger χ value is, the more sensitive weight changes. ξ_0 is the critical point of the psychological preference factor, when $\xi_k = \xi_0$, the value of the psychological preference function is $\frac{\omega_j^k}{2}(j = 1, 2, \dots, n, k = 1, 2, \dots, s)$, indicating that the influence of psychological preference on the weight is at a medium level at this time.

Theorem 1. For any DM $d_k(k = 1, 2, \dots, s)$, if the psychological preference factor is $\xi_k(k = 1, 2, \dots, s)$, then for any $\xi_k > \xi_{k'}(k \neq k')$, there is $\omega_j^k(\xi_k) \leq \omega_j^{k'}(\xi_{k'})$.

Proof. The proof is provided in Appendix A.1.

Corollary 1. For any DM $d_k(k = 1, 2, \dots, s)$, the psychological preference function $f(\xi_k) = \frac{1}{1 + e^{-\chi(\xi_k - \xi_0)}}$ is convex, and the weight of attribute $\xi_k(k = 1, 2, \dots, s)$ can adapt to different decision environments with different values of $\omega_j^k(\xi_k)(j = 1, 2, \dots, n, k = 1, 2, \dots, s)$.

Proof. The proof is provided in Appendix A.2.

4.1.2. The determination of the weights of attributes integrating cooperative contribution and psychological preference

In complex and uncertain group decision-making environments, decision makers (DMs) often exhibit cognitive heterogeneity and goal conflicts. To structurally and quantitatively characterize the contribution of each DM, this paper adopts the Shapley value from cooperative game theory to represent their marginal contribution under coalition formation. The Shapley value is defined as follows:

$$\phi_k = \sum_{S \subseteq D \setminus \{d_k\}} \frac{|S|!(|D| - |S| - 1)!}{|D|!} [V(S \cup \{d_k\}) - V(S)], k = 1, 2, \dots, s. \quad (6)$$

Unlike the traditional approach of using Shapley values as post-event evaluation or contribution allocation indicators (Meng et al., 2023a; Wu et al., 2023a), this paper directly introduces the marginal contribution of Shapley into the process of determining attribute weights. Specifically, the marginal contribution in this paper is not calculated based on fixed or exogenous given alliance returns, but is endogenous generated through the alliance feature function influenced by attribute weights adjusted by psychological preferences, thereby enabling psychological preferences to participate in the generation process of Shapley's marginal contribution.

On this basis, considering the attribute weights adjusted by psychological preferences and the marginal contribution in the cooperative game, this paper further constructs an attribute weight optimization model to realize the adaptive response of weight distribution to the psychological state of decision makers and the cooperative contribution. The optimization model (7) is as follows:

$$\begin{aligned} M_1 : \max & \sum_{k=1}^s \sum_{S \subseteq D \setminus \{d_k\}} \frac{|S|!(|D| - |S| - 1)!}{|D|!} [V(S \cup \{d_k\}) - V(S)] \\ & V(S) = \sum_{\{d_k\} \in S} \sum_{i=1}^m \sum_{j=1}^n \omega_j^k(\xi_k) v_{ij}^k \pi_{ij}^k(\omega_j^k(\xi_k)) \quad (7-1) \\ & v_{ij}^k = \begin{cases} (z_{ij}^k - \bar{z}_j^k)^{\hat{\alpha}}, & z_{ij}^k \geq \bar{z}_j^k \\ -\lambda(\bar{z}_j^k - z_{ij}^k)^{\hat{\beta}}, & z_{ij}^k < \bar{z}_j^k \end{cases} \quad (7-2) \\ & \pi_{ij}^k(\omega_j^k) = \begin{cases} \frac{(\omega_j^k(\xi_k))^{\gamma}}{[(\omega_j^k(\xi_k))^{\gamma} + (\omega_j^k(\xi_k))^{\delta}]^{1/\gamma}}, & z_{ij}^k \geq \bar{z}_j^k \\ \frac{(\omega_j^k(\xi_k))^{\delta}}{[(\omega_j^k(\xi_k))^{\delta} + (1 - \omega_j^k(\xi_k))^{\delta}]^{1/\delta}}, & z_{ij}^k < \bar{z}_j^k \end{cases} \quad (7-3) \\ & s.t. \begin{cases} z_{ij}^k = \frac{1}{m} \sum_{i=1}^m z_{ij}^k \quad (7-4) \\ \sum_{j=1}^n \omega_j^k(\xi_k) = 1 \quad (7-5) \\ \sum_{j=1}^n \omega_j^k = 1 \quad (7-6) \\ \omega_j^k(\xi_k) = \frac{\omega_j^k}{1 + e^{-\chi(\xi_k - \xi_0)}} \quad (7-7) \\ \chi \geq 0, \xi_k \in [0, 1], \omega_j^k \geq 0, \omega_j^k(\xi_k) \geq 0, 0 < \hat{\alpha}, \hat{\beta} < 1, \lambda > 1 \quad (7-8) \\ i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s \quad (7-9) \end{cases} \end{aligned} \quad (7)$$

where D denotes the set of DMs, and S refers to a subset of D that excludes the DM $d_k(k = 1, 2, \dots, s)$.

Since the decision variables in model (7) only include the attribute weights ω_j^k and their psychological preference mappings $\omega_j^k(\xi_k)$, each evaluation requires a weighted traversal over all combinations of DMs, alternatives, and attributes (s, m, n) to compute the expected utility and marginal contribution. Therefore, the time complexity is $O(smn)$.

The aim of the model's objective function is to enhance the total utility for every DM in a cooperative setting. This ensures that the alternative exhibiting the greatest overall utility is chosen through a multi-attribute assessment. By maximizing this objective function, the model accounts for the preferences and judgments of multiple DMs, optimizes the efficiency of the entire rescue system, and ensures that the decision outcomes are both practical and scientifically sound with multi-party participation. Constraint (7-1) defines how to evaluate the effectiveness of cooperation among DMs by measuring the overall utility of decision

groups. This constraint ensures that rescue resources are optimally allocated under multi-party cooperation, thereby improving overall rescue efficiency. Constraints (7-2) and (7-3) define the value function and weight function from prospect theory, while constraint (7-4) calculates the reference points in prospect theory. Constraints (7-5) and (7-6) ensure that the sum of attribute weights equals 1. Constraint (7-7) describes how the DM's psychological preference factor ζ_k ($k = 1, 2, \dots, s$) influences the weight of each attribute, reflecting the nonlinear regulatory effect of a DM's psychological state on decision-making behavior in high-pressure environments. Other constraints set the allowable ranges for parameter values.

Theorem 2. The model (7) has an optimal solution.

Proof. The proof is provided in [Appendix A.3](#).

4.2. Trust relationship representation based on quantum intuitionistic fuzzy set

In MAGDM, trust relationships among decision makers exhibit pronounced dynamic evolution and are often accompanied by interference effects induced by interactions. Such trust states typically involve not only trust and distrust but also hesitation and uncertainty, and these components may change continuously with group interactions and decision feedback. Although quantum-like Bayesian networks can describe trust propagation and belief updating processes based on quantum probability theory, their modeling generally remains at the level of probability distributions, making it difficult to capture the internal structural differentiation of multiple cognitive components within a single trust relationship and their dynamic coupling relationships ([Li et al., 2026](#); [Wang et al., 2024](#)).

Existing studies, such as [Reiser et al. \(2016\)](#), have explored quantum representations and operational interpretations of intuitionistic fuzzy information from the perspective of quantum computing. However, in trust-network-based group decision-making, a more critical issue lies in providing a unified and evolvable state representation for trust, distrust, and hesitation. Motivated by this requirement, this paper introduces a quantum-state-based representation and extends intuitionistic fuzzy sets to quantum intuitionistic fuzzy sets (QIFS). By embedding trust, distrust, and hesitation into different basis-state components of a quantum state, QIFS enables the representation of the multi-dimensional cognitive structure of trust relationships within a unified Hilbert space. Moreover, owing to the normalization constraint of quantum states, QIFS preserves the invariance of overall uncertainty during quantum state evolution ([Yang et al., 2025](#)), thereby achieving a balance between the dynamic expression of trust relationships and structural stability.

Definition 2. Let X be a non-empty set, $X = \{x_1, x_2, \dots, x_n\}$, and the QIFS Q is defined by a mapping $f : X \rightarrow [0, 1] \times [0, 1] \times [0, 1]$, as follows $Q(x) = \{ \langle \mu_Q(x), \nu_Q(x), \pi_Q(x) \rangle | x \in X \}$.

where for $\forall x \in X$, $\mu_Q(x) + \nu_Q(x) + \pi_Q(x) = 1$ is satisfied. The representation of element x in the quantum state $|\psi_x\rangle$ is:

$$|\psi_x\rangle = \alpha_x|0\rangle + \beta_x|1\rangle + \gamma_x|h\rangle. \quad (9)$$

where $|0\rangle$, $|1\rangle$ and $|h\rangle$ are the orthogonal ground states defined on the Hilbert space $\mathcal{H}$, corresponding to membership, non-membership and hesitancy respectively. The probability amplitudes α_x, β_x , and γ_x are complex and satisfy the normalization condition: $|\alpha_x|^2 + |\beta_x|^2 + |\gamma_x|^2 = 1$.

In quantum mechanics, the inner product of quantum states is used to measure the degree of similarity between two quantum states, as it reflects the overlap of states and the probability of their evolution. In this paper, this property of the inner product is employed to measure the similarity between QIFS. The similarity is obtained by taking the modulus square of the inner product, which is then converted into a distance measure. By using this approach, the complex relationships between quantum states can be quantified as a distance measure in an intuitive manner, while preserving the precision of quantum mechanics. This allows for a more accurate representation of the correlation between QIFS.

Definition 3. Suppose there are two QIFS $Q_1(x)$ and $Q_2(x)$, whose corresponding quantum states are $|\psi_{x_1}\rangle$ and $|\psi_{x_2}\rangle$, respectively. Then the similarity of two QIFS is defined as:

$$S(Q_1(x), Q_2(x)) = \left| \langle \psi_{x_1} | \psi_{x_2} \rangle \right|^2 = \left| \alpha_{x_1}^* \alpha_{x_2} + \beta_{x_1}^* \beta_{x_2} + \gamma_{x_1}^* \gamma_{x_2} \right|^2. \quad (10)$$

where $\alpha_{x_1}^*$, $\beta_{x_1}^*$, and $\gamma_{x_1}^*$ are complex conjugations of α_{x_1} , β_{x_1} , and γ_{x_1} , respectively.

On this basis, the distance measure of two QIFS is defined as:

$$d(Q_1(x), Q_2(x)) = \sqrt{1 - S(Q_1(x), Q_2(x))}. \quad (11)$$

Theorem 3. For any two QIFS $Q_1(x)$ and $Q_2(x)$, the distance measure needs to satisfy three properties:

- 1) $0 \leq d(Q_1(x), Q_2(x)) \leq 1$.
- 2) $d(Q_1(x), Q_2(x)) = d(Q_2(x), Q_1(x))$.
- 3) If $Q_1(x) = Q_2(x)$, then $d(Q_1(x), Q_2(x)) = 0$.

Remark 1. Properties 2 and 3 can be directly verified from the definition of the distance function, and thus only the proof of Property 1 is provided here.

Proof. The proof is provided in [Appendix A.4](#).

Definition 4. Suppose there are two QIFS $Q_1(x)$ and $Q_2(x)$, whose corresponding quantum states are $|\psi_{x_1}\rangle$ and $|\psi_{x_2}\rangle$. Then the addition operation of two QIFS is defined as:

$$Q_1 + Q_2 = \left\langle \frac{\sum_{i=1}^2 |\alpha_i|^2}{\sum_{i=1}^2 (|\alpha_i|^2 + |\beta_i|^2 + |\gamma_i|^2)}, \frac{\sum_{i=1}^2 |\beta_i|^2}{\sum_{i=1}^2 (|\alpha_i|^2 + |\beta_i|^2 + |\gamma_i|^2)}, \frac{\sum_{i=1}^2 |\gamma_i|^2}{\sum_{i=1}^2 (|\alpha_i|^2 + |\beta_i|^2 + |\gamma_i|^2)} \right\rangle. \quad (12)$$

On the basis, this paper constructs QIFS to characterize the trust relationships among DMs. Within this framework, the membership degree denotes the level of trust, while the non-membership degree reflects the level of distrust. The adoption of QIFS allows for a more precise description of ambiguity and uncertainty in trust relationships, particularly in situations where trust and distrust coexist simultaneously.

Remark 2. The addition operation defined in Definition 4 is designed to ensure the mathematical validity and semantic consistency of QIFS aggregation. Specifically, it follows the principle of quantum-state superposition, where the squared probability amplitudes of trust, distrust, and hesitation are accumulated and normalized to preserve the condition $|\alpha_x|^2 + |\beta_x|^2 + |\gamma_x|^2 = 1$.

This guarantees that the resulting triplet remains within the valid QIFS space while maintaining probability conservation. By aggregating the squared amplitudes rather than complex amplitudes, the operation avoids phase-induced interference and yields a stable and interpretable result. From a fuzzy-logic viewpoint, this normalized form retains the semantic integrity of partial trust and ensures that the fusion of multiple trust states remains additive, coherent, and consistent with the underlying quantum-fuzzy framework.

4.3. A trust correction model considering marginal contribution based on QIFS representing trust relationships

In social networks, the formation and evolution of trust relationships exert a significant impact on collective decision outcomes. To formally characterize these relationships, this paper constructs QIFS, which exploit the superposition property of quantum states to reflect the co-existence of fuzziness and uncertainty. Within this representation, the membership degree corresponds to trust, the non-membership degree corresponds to distrust, and the hesitancy degree quantifies the indeterminacy inherent in evaluating interpersonal trust. Accordingly, for any pair of DMs d_k and d_h in the network, the trust relation from d_k to d_h can be expressed in the form of a QIFS, as follows:

$$|\psi_{kh}\rangle = \alpha_{kh}|0\rangle + \beta_{kh}|1\rangle + \gamma_{kh}|h\rangle, k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h. \quad (13)$$

Therefore, the trust matrix in a social network is $T = (t_{kh})_{s \times s} = (|\psi_{kh}\rangle)_{s \times s}$, $t_{kh} \in [0, 1]$. Based on the initial trust matrix $T = (t_{kh})_{s \times s}$, the degree centrality of each DM is calculated. The comprehensive trust level CTL_k of DM d_k is defined as:

$$CTL_k = \frac{d\left(\sum_{h=1, h \neq k}^s |\psi_{kh}\rangle, \sum_{h=1, h \neq k}^s |\Psi_{hk}\rangle\right)}{\max d\left(\sum_{h=1, h \neq k}^s |\psi_{kh}\rangle, \sum_{h=1, h \neq k}^s |\Psi_{hk}\rangle\right)}, k = 1, 2, \dots, s. \quad (14)$$

where, $|\psi_{kh}\rangle (k = 1, 2, \dots, s, h = 1, 2, \dots, s, h \neq k)$ represents the quantum state trust out degree, $|\Psi_{hk}\rangle (k = 1, 2, \dots, s, h = 1, 2, \dots, s, h \neq k)$ represents the quantum state trust in degree.

In practical group decision-making, apart from individuals' direct contributions to evaluation tasks, interpersonal trust and interactive relationships also have a profound impact on actual influence and weight assignment. Trust relationships are mainly used to construct the information adoption structure within the group, reflecting whose opinions are more highly recognized by the group, thereby influencing the generation mechanism of the overall decision-making ranking results. Considering that traditional methods often fall short in capturing the uncertainty and dynamic evolution of trust relationships, this paper proposes a QIFS to describe trust in a dynamic and fuzzy manner. On this basis, a two-stage trust modeling framework is proposed to dynamically reflect the structure of interpersonal trust and its optimized adjustment, as shown in Algorithm 1.

Algorithm 1 A two-stage dynamic trust correction framework based on quantum intuitionistic fuzzy set representation.

- 1: **Input:** Initial trust matrix $T = (t_{kh})_{s \times s}$; marginal contribution vector $\phi = [\phi_1, \phi_2, \dots, \phi_s]^T$ of DMs.
- 2: **Output:** Updated comprehensive trust levels CTL_k .
- 3: **Begin**
- 4: **Stage 1: Initial trust level computation**
- 5: **Step 1:** Collect pairwise trust data among all DMs and normalize it into intuitionistic fuzzy triplets $(\mu_{kh}, \nu_{kh}, \pi_{kh})$.
- 6: **Step 2:** According to Eq. (13), map each triplet $(\mu_{kh}, \nu_{kh}, \pi_{kh})$ into a quantum state.
- 7: **Step 3:** Compute the comprehensive trust level of each DM based on Eq. (14).
- 8: **Stage 2: Trust revision and dynamic adjustment**
- 9: **Step 4:** Based on Model (15), integrate each CTL_k with its corresponding marginal contribution ϕ_k to obtain the revised comprehensive trust level CTL_k .
- 10: **Step 5:** Output the updated comprehensive trust levels CTL_k .
- 11: **End**

This two-stage process comprehensively accounts for the dynamic relationship between contribution and trust, leading to a more reasonable allocation of DM weights and ultimately improving the quality and fairness of group decision-making. The optimization model is as follows:

$$M_2 : \max \sum_{k=1}^s CTL_k$$

$$\left\{ \begin{array}{l} \sum_{k=1}^s \lambda_k = 1 \quad (15-1) \\ \left| \frac{\lambda_k}{\lambda_h} - \frac{\phi_k}{\phi_h} \right| \leq \epsilon \quad (15-2) \\ \lambda_k \geq \eta \frac{CTL_k}{\sum_{k=1}^s CTL_k} \quad (15-3) \\ CTL_k = 1 + \beta \cdot CTL_k \cdot \ln(1 + \alpha \phi_k) \quad (15-4) \\ \lambda_k = \frac{CTL_k \cdot \phi_k}{\sum_{k=1}^s CTL_k \cdot \phi_k} \quad (15-5) \\ \lambda_k \geq 0 \quad (15-6) \\ \alpha \in [0, 1], \beta \in [0, 1], \eta \in [0, 1], \epsilon \geq 0 \quad (15-7) \\ h = 1, 2, \dots, s, k = 1, 2, \dots, s \quad (15-8) \end{array} \right. \quad (15)$$

where the decision variables are CTL_k and λ_k , both of them can be obtained through closed-form expressions. Specifically, each CTL_k and λ_k can be computed in constant time, and the overall computation only requires a single linear traversal for $k = 1, \dots, s$, resulting in a total time complexity of $O(s)$.

$\phi_k (k = 1, 2, \dots, s)$, $\phi_h (h = 1, 2, \dots, s)$, $CTL_k (k = 1, 2, \dots, s)$ can be calculated in the first stage. ϵ is the permissible deviation in the fairness constraint based on the Shapley value, it determines the allowable error between the weight ratio and the Shapley value ratio. $\alpha (\alpha \in [0, 1])$ is a factor used to control the strength of the Shapley value's influence in modifying the trust level. $\beta (\beta \in [0, 1])$ is an influence coefficient that represents the effect of the initial trust level on the final trust level. It controls the effect of the modified trust level on the weight allocation. η is a scale factor in the constraint of the relationship between trust level and weight, which is used to ensure that the modified trust level can be reasonably reflected in the weight.

In this model, the objective function is designed to optimize the weight allocation of DMs by adjusting their trust levels. Specifically, the goal is to maximize the combined influence of each DM. The constraint condition (15-1) ensures that the weight of each DM satisfies the normalization condition. Constraint (15-2) imposes a fairness constraint based on the Shapley value, limiting the difference between the weight ratio and the Shapley value ratio. This ensures that the weight distribution reflects the actual contribution of the DMs and prevents unfair weight allocation in extreme cases. Constraints (15-3) prevent inconsistencies between trust levels and weights. Constraint condition (15-4) provides the revised trust level calculation formula, which takes the form of a logarithmic function. This approach mitigates the influence of the Shapley value on trust levels, ensuring that even if the Shapley value is large, its impact on trust levels is not disproportionately exaggerated. Constraint (15-5) defines the formula for calculating the weight of each DM. The remaining constraints specify the allowable range for the value of each parameter.

Theorem 4. The model (15) has an optimal solution.

Proof. Since the proof follows the same line of reasoning as Theorem 2, it is not repeated here.

4.4. Interference angle measure based on KL divergence

The KL divergence can effectively describe the degree of information deviation and separability between probability distributions, and is often used to measure the differences and discriminability between subjective distributions or feature-based distributions (Al-Jarrah et al., 2023; Zhang et al., 2023). From the perspective of probability mapping, in this paper, the normalized preference vector of each DM is regarded as a subjective probability distribution defined in the alternative space, thereby enabling the preference differences among different DMs to be compared within a unified probability representation framework.

On this basis, this paper proposes the symmetrical KL divergence to describe the deviation degree of the subjective probability distribution among DM, providing a measurement basis with information theory significance for cognitive differences. In addition, the KL divergence is not used as the final difference evaluation index, but rather as an important intermediate quantity for the subsequent construction of the interference angle, to support the angular expression of cognitive interference intensity.

Firstly, the DM preference vector $\vartheta_k = (\vartheta_{k1}, \vartheta_{k2}, \dots, \vartheta_{km})^T$, $(k = 1, 2, \dots, s)$ is constructed.

$$\vartheta_k = \begin{bmatrix} p(d_k)p(a_1|d_k) \\ p(d_k)p(a_2|d_k) \\ \vdots \\ p(d_k)p(a_m|d_k) \end{bmatrix}_{m \times 1}, k = 1, 2, \dots, s. \quad (16)$$

On this basis, the DM preference matrix $X = (x_{ik})_{m \times s}$ is constructed, where each row corresponds to the preference vector of a DM, that is, $x_{ik} = p(d_k)p(a_i|d_k)$.

KL divergence $D_{KL}(d_k \| d_h)$ ($k = 1, 2, \dots, s, h = 1, 2, \dots, s, h \neq k$) measures the loss of information from the scoring distribution of DM d_k ($k = 1, 2, \dots, s$) relative to the scoring distribution of DM d_h ($h = 1, 2, \dots, s$). The KL divergence between two DMs is calculated as follows:

$$D_{KL}(d_k \| d_h) = \sum_{i=1}^m x_{ik} \ln \frac{x_{ik}}{x_{ih}}, k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h. \quad (17)$$

Secondly, since KL divergence is asymmetric, it is necessary to calculate the bidirectional KL divergence. By averaging the two directions, a symmetric KL divergence is obtained, which eliminates the influence of directionality. Symmetrizing the KL divergence makes the measurement more balanced and fair.

$$D'_{KL}(d_k \| d_h) = \frac{1}{2} (D_{KL}(d_k \| d_h) + D_{KL}(d_h \| d_k)), \quad (18)$$

$$k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h.$$

Finally, the symmetric KL divergence is mapped to a range of 0 to 180 degrees, which is converted into the interference angle.

$$\theta_{kh} = \frac{D'_{KL}(d_k \| d_h)}{\max D'_{KL}(d_k \| d_h)} \times 180^\circ, k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h. \quad (19)$$

4.5. Quantum probability computation for alternative

In meteorological emergency decision-making problems, the high complexity and rapid variability of weather conditions, as well as the interaction among multiple DMs, make the decision-making process simultaneously affected by uncertainty and cognitive interference. To characterize the evaluation information, trust relationships and mutual interference effects of multiple DMs under a unified framework, this paper, based on the aforementioned trust correction, psychological preference adjustment and attribute weight determination, processes the interference effects among DMs based on QLBN.

The QLBN constructed in this paper consists of a three-layer structure. The first layer is the DM layer, which is used to represent each DM and their corresponding prior probability. This prior probability is determined by the DM's weight and is used to reflect their credibility and influence within the group. The second layer is the alternative scheme layer, which links the evaluation behavior of DMs with specific rescue schemes through conditional probability. The conditional probability is generated from multi-attribute evaluation information regulated by psychological preferences. The third layer is the environmental factor layer, which is used to describe the potential impact of external uncertain factors such as meteorological changes on the decision-making process, as shown in Fig. 3.

Unlike most existing QLBNS that regard prior probability and conditional probability as exogenous given, the QLBN constructed in this paper links the key probability nodes in the network with trust relationships, psychological preferences, and multi-attribute evaluation mechanisms, making it an endogenous expression of group decision-making behavior rather than merely a probability propagation tool for interference computations. The QLBN constructed in this paper is as follows:

Firstly, make the DM weight the first layer probability of the QLBN:

$$p(d_k) = \lambda_k, k = 1, 2, \dots, s. \quad (20)$$

Secondly, combined with decision units and attributes' weights, it serves as a tool to link decision groups with index attributes in the QLBN.

$$p(a_i | d_k) = \frac{\sum_{j=1}^n \varpi_j^k(\xi_k) z_{ij}^k}{\sum_{k=1}^s \sum_{j=1}^n \varpi_j^k(\xi_k) z_{ij}^k}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (21)$$

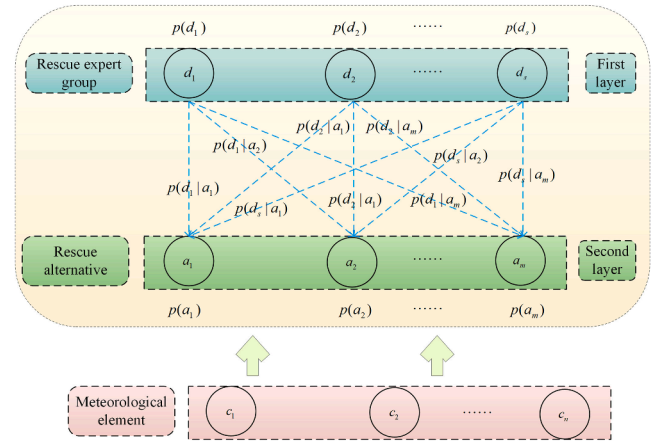


Fig. 3. Quantum-like Bayesian network based on meteorological disaster.

Finally, the quantum probability of each alternative is calculated and ranked.

$$p(a_i) = \sigma \left| \sum_{k=1}^s \varphi_{d_k} e^{i\theta_{d_k}} \varphi_{a_i | d_k} e^{i\theta_{d_h}} \right|^2$$

$$= \sigma \left[\sum_{k=1}^s |\varphi_{d_k} \varphi_{a_i | d_k}|^2 + 2 \sum_{k=1}^{s-1} \sum_{h=k+1}^s \varphi_{d_k} \varphi_{a_i | d_k} \varphi_{d_h} \varphi_{a_i | d_h} \cos(\theta_{kh}) \right], \quad (22)$$

$$i = 1, 2, \dots, m.$$

where,

$$\sigma = 1 / \sum_{i=1}^m \left| \sum_{k=1}^s \varphi_{d_k} \varphi_{a_i | d_k} \right|^2. \quad (23)$$

4.6. Decision steps

This paper proposes a dynamic trust network group decision-making method based on cooperative game theory and quantum probability theory. This section summarizes the execution procedure of the proposed group decision-making framework by organizing the previously introduced model components into a coherent decision process. Fig. 4 illustrates the logical relationships among the different modules, including information preprocessing, psychological preference-based weight determination, dynamic trust updating, and quantum-like probabilistic inference, thereby clarifying the data flow, parameter interactions, and output generation within the overall framework. The corresponding decision steps are outlined as follows:

Step 1: Gather the preliminary assessment information matrix for each DM, then standardize the initial decision matrix of all DMs in line with Eq. (4), resulting in the normalized decision matrix for the DMs.

Step 2: Construct a psychological preference function to describe individual risk perception and behavioral tendencies. On this basis, the Shapley value (Eq. (6)) is applied to measure each DM's marginal contribution, and the optimization Model (7) is formulated to determine the attribute weights $\varpi_j^k(\xi_k)$, which serve as an input variable for Model (15) in Step 3.

Step 3: Use the QIFS to represent the dynamic and fuzzy trust relationships among DMs. Then, a two-stage dynamic trust-revision framework is established, and Model (15) is solved to update the trust levels and obtain the DMs' weights λ_k .

Step 4: Construct a QLBN, where the attributes' weights obtained in Step 2 are used to compute conditional probabilities $p(a_i | d_k)$ (Eq. (21)) and the DM weights from Step 3 are used to calculate prior probabilities $p(d_k)$ (Eq. (20)). Additionally, the KL divergence is used to measure preference distribution differences among DMs and map them to interference angles by Eqs. (16–19).

Step 5: Based on the prior probabilities $p(d_k)$ and conditional probabilities $p(a_i|d_k)$ obtained in Step 4, compute the quantum probabilities of alternatives using Eq. (22) and rank them.

5. Case analysis

5.1. Background

On April 12, 2024, a severe forest fire occurred in Kunming City, Yunnan Province, China. The fire spread rapidly, causing extensive destruction of forest resources and threatening nearby villages, transportation routes, and power and communication facilities (<https://xinwen.bjd.com.cn/content/s661a9fe5e4b0fa9ed6b8e6cc.html>). The extreme heat and heavy smoke made evacuation and firefighting efforts extremely difficult. The fire's development was further complicated by complex terrain and shifting wind directions, increasing the risk of secondary fires. In response to this urgent and complex situation, the Kunming emergency management department decided to adopt a MAGDM approach, integrating the opinions of DMs from various fields and stakeholders to scientifically determine the optimal firefighting and rescue strategy.

The decision-making group consists of five DMs. The background of DMS is provided in the Appendix B. When formulating a response strategy, the DMs considered five main attributes: 1) Response Speed c_1 : Measures the timeliness of rescue forces and command arriving at the fire scene. The scoring range is [1,10], where 1 indicates very slow response, leading to loss of fire control, and 10 indicates extremely rapid response, effectively controlling the fire. 2) Resource Allocation Efficiency c_2 : Reflects the efficiency of distributing manpower, equipment, and materials across different fire areas. The scoring range is [1,10], with 1 indicating chaotic allocation and 10 indicating scientific and efficient resource utilization. 3) Fire Controllability c_3 : Evaluates the effectiveness of measures in controlling fire spread and reducing the risk of secondary fires. The scoring range is [1,10], with 1 representing difficulty in controlling the fire and 10 representing highly controllable fire conditions. 4) Ecological Impact c_4 : Measures the impact of firefighting and rescue actions on the surrounding ecological environment. The scoring range is [1,10], where 1 indicates severe ecological damage and 10 indicates well-protected ecosystems. 5) Cost Control c_5 : Evaluates the rationality and efficiency of financial and material expenditures. The scoring range is [1,10], with 1 indicating serious overspending and waste and 10 indicating excellent cost control and efficient use of funds. Furthermore, all these five decision-making attributes are benefit attributes.

In response to the disaster, the decision team developed five alternatives, each representing a different approach to resource allocation and prioritization:

1) Alternative a_1 : Prioritize the safety of rescue personnel and those being rescued by adopting a more cautious rescue strategy to avoid potential secondary disasters. This alternative may slow down the response speed of firefighting operations but can effectively reduce casualties and accident risks. 2) Alternative a_2 : Deploy firefighting forces in a distributed manner across multiple areas to ensure that fire in different regions is controlled, avoiding over-concentration of resources in a single area. Although the intensity of operations in individual areas may be relatively lower, overall fire controllability is improved. 3) Alternative a_3 : Under a limited budget, optimize existing resource allocation and control costs reasonably, balancing firefighting actions with ecological protection to avoid waste. However, strict cost control may restrict the scale and coverage of rescue actions in the short term. 4) Alternative a_4 : Concentrate all firefighting forces on the most severe fire areas to rapidly suppress major flames and prevent further spread. While this centralized strategy ensures decisive control of high-risk zones, it may temporarily reduce resource coverage in peripheral areas, requiring subsequent coordinated reinforcement. 5) Alternative a_5 : Focus resources on establishing firebreaks and continuous monitoring systems, providing strong long-term fire control and secondary fire prevention capacity.

However, this alternative may not achieve rapid fire suppression in the early stages and is more suitable for long-term prevention and control efforts.

5.2. Decision steps

Step 1: Collect the initial evaluation information matrix $\hat{Z}^k (k = 1, 2, \dots, 5)$ for DMs, as shown in Table 2.

The normalization process is carried out according to Eq. (4), and the normalized decision matrix of each DM is obtained, as shown in Table 3.

Step 2: Based on the normalized decision matrix, solve the model (7) to obtain the optimal attribute weights of each DM, as shown in Table 4.

Step 3: Firstly, the representation matrix of trust relations based on QIFS is obtained, as shown in Table 5.

Secondly, the comprehensive trust degree of each DM is calculated according to Eq. (14) as follows:

$$CTL_1 = 0.4344, CTL_2 = 1.0000, CTL_3 = 0.4062, \\ CTL_4 = 0.4956, CTL_5 = 0.2458.$$

On this basis, the weight of each DM is obtained by solving model (15):

$$\Lambda = \{0.1912, 0.2543, 0.1840, 0.1986, 0.1719\}.$$

Step 4: Construct a QLBN. The first layer is the DM's weights, and the conditional probability is calculated according to Eq. (21), as shown in Table 6.

On this basis, the interference angle between DMs is calculated according to Eqs. (16–19).

$$\theta_{12} = 26.2840^\circ, \theta_{13} = 101.0233^\circ, \theta_{14} = 138.8523^\circ, \\ \theta_{15} = 0.5883^\circ, \theta_{23} = 78.9190^\circ, \theta_{24} = 74.6819^\circ, \\ \theta_{25} = 24.6642^\circ, \theta_{34} = 180^\circ, \theta_{35} = 107.1596^\circ, \\ \theta_{45} = 132.6520^\circ.$$

Step 5: Calculate the quantum probability values of each alternative:

$$p(a_1) = 0.1674, p(a_2) = 0.2177, p(a_3) = 0.1325, \\ p(a_4) = 0.2782, p(a_5) = 0.2042.$$

Therefore, the ranking result of the alternatives is: $a_4 > a_2 > a_5 > a_1 > a_3$. This result indicates that in high-risk and high-time-pressure disaster scenarios, concentrating resources to quickly extinguish the main fire area (alternative a_4) is the optimal solution, which is in line with the actual emergency management strategy of "controlling the main fire first and then stabilizing the overall situation". a_2 ranks second, reflecting the group's high recognition of the "multi-point defense and overall control" strategy under the influence of the coupling mechanism of psychological preferences and trust. Although a_5 has insufficient short-term control power, it makes a relatively high contribution in long-term defense and ranks in the middle. The relatively low rankings of a_1 and a_3 indicate that under conditions of sudden disasters, the group is more inclined to prioritize ensuring response speed and control capabilities rather than minimizing costs or avoiding risks.

5.3. Sensitivity analysis

According to group decision-making theory, the core purpose of sensitivity analysis is to test the stability, robustness and behavioral interpretability of the model under changes in key parameters (Ji et al., 2025). Since the multi-attribute quantum group decision-making model proposed in this paper integrates the evolution of psychological preferences and trust correction mechanisms, both of these dynamic factors will affect attribute weights, marginal contributions, and decision ranking results through parameter perturbations. Therefore, it is necessary to conduct a systematic analysis of parameter sensitivity.

1) The influence of psychological preference factors $\xi_k (k = 1, 2, \dots, 5)$ on the weight of DMs' attributes

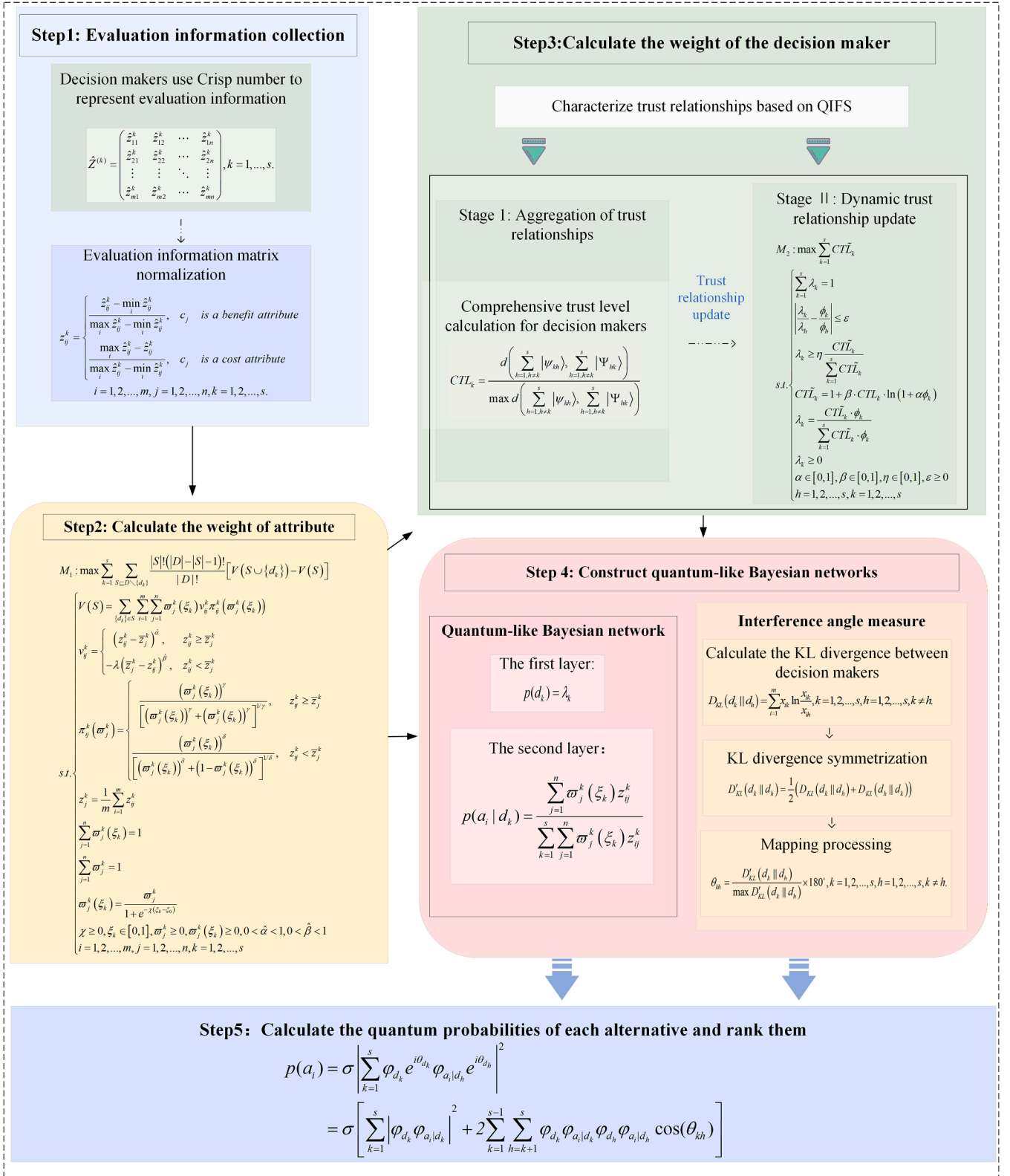


Fig. 4. Decision steps of the proposed model.

Table 2
Initial evaluation information matrix $\hat{Z}^k (k = 1, 2, \dots, 5)$.

	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5
	d_1					d_2					d_3				
a_1	7	4	8	5	7	1	10	3	7	4	3	1	4	2	8
a_2	10	3	7	8	5	9	3	5	3	7	4	2	6	6	10
a_3	4	8	8	3	6	5	9	7	2	4	4	6	2	10	2
a_4	5	2	8	6	2	9	2	10	9	10	10	4	8	7	9
a_5	5	1	10	6	9	5	2	4	7	8	8	5	2	5	8
	d_4					d_5									
a_1	10	9	9	1	9	2	1	7	7	8					
a_2	7	9	8	1	8	5	3	8	6	3					
a_3	8	3	1	8	3	1	3	5	3	1					
a_4	3	1	5	10	7	5	10	7	7	9					
a_5	10	9	7	9	8	10	10	3	7	1					

Table 3
Normalized decision matrix $Z^k (k = 1, 2, \dots, 5)$.

	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5
	d_1					d_2				
a_1	0.5000	0.4286	0.3333	0.4000	0.7143	0.0000	1.0000	0.0000	0.7143	0.0000
a_2	1.0000	0.2857	0.0000	1.0000	0.4286	1.0000	0.1250	0.2857	0.1429	0.5000
a_3	0.0000	1.0000	0.3333	0.0000	0.5714	0.5000	0.8750	0.5714	0.0000	0.0000
a_4	0.1667	0.1429	0.3333	0.6000	0.0000	1.0000	0.0000	1.0000	1.0000	1.0000
a_5	0.1667	0.0000	1.0000	0.6000	1.0000	0.5000	0.0000	0.1429	0.7143	0.6667
	d_3					d_4				
a_1	0.0000	0.0000	0.3333	0.0000	0.7500	1.0000	1.0000	1.0000	0.0000	1.0000
a_2	0.1429	0.2000	0.6667	0.5000	1.0000	0.5714	1.0000	0.8750	0.0000	0.8333
a_3	0.1429	1.0000	1.0000	1.0000	0.0000	0.7143	0.2500	0.0000	0.7778	0.0000
a_4	1.0000	0.6000	1.0000	0.6250	0.8750	1.0000	0.0000	0.5000	1.0000	0.6667
a_5	0.7143	0.8000	0.0000	0.3750	0.7500	1.0000	1.0000	0.7500	0.8889	0.8333
	d_5									
a_1	0.1111	1.0000	0.8000	1.0000	0.8750					
a_2	0.4444	0.2222	1.0000	0.7500	0.2500					
a_3	0.0000	0.2222	0.0000	0.5000	0.2500					
a_4	0.4444	1.0000	0.8000	1.0000	0.8750					
a_5	1.0000	1.0000	1.0000	1.0000	1.0000					

Table 4
The weights of the optimal attributes $\varpi_j^k(\xi_k) (j = 1, 2, \dots, 5, k = 1, 2, \dots, 5)$ of each DM.

	c_1	c_2	c_3	c_4	c_5
d_1	0.2119	0.1795	0.2234	0.2446	0.1406
d_2	0.3027	0.2425	0.0742	0.1651	0.2155
d_3	0.1427	0.0685	0.3851	0.1045	0.2992
d_4	0.1634	0.2077	0.1909	0.2129	0.2251
d_5	0.2480	0.1906	0.1964	0.2383	0.1267

From the perspective of the psychological preference dimension, the psychological preference function (Eq. (5)) in model (7) establishes a nonlinear mapping relationship between psychological states and attribute weights. When the psychological preference factor $\xi_k (k = 1, 2, \dots, 5)$ changes, the degree of attention DMs pay to different attributes will undergo a dynamic transformation, as shown in Fig. 5.

Specifically, Fig. 5 illustrates the changing emphasis DMs place on each attribute as the psychological preference factor $\xi_k (k = 1, 2, \dots, 5)$ varies between 0 and 1. As $\xi_k (k = 1, 2, \dots, 5)$ increases, some attributes show a gradual rise in weight, while others experience a decline. This shift indicates that DMs' focus can significantly change depending on their psychological state, especially during different emergency situations.

The psychological preference factor reflects how DMs respond to risks and uncertainties. At lower values of $\xi_k (k = 1, 2, \dots, 5)$, DMs tend to

be more conservative or cautious, resulting in lower weights for certain key attributes. Conversely, at higher values of $\xi_k (k = 1, 2, \dots, 5)$, DMs exhibit greater confidence, assigning higher weights to specific critical attributes.

Therefore, in varying disaster scenarios, the psychological state of DMs directly influences their evaluation of different rescue measures, subsequently affecting the selection of the final alternative. This analysis not only supports the validity of the model, but also underscores the importance of incorporating psychological preference factors in uncertain environments to enhance the accuracy of the decision-making process.

2) The influence of psychological preference factors $\xi_k (k = 1, 2, \dots, 5)$ on the contribution degree of DMs

Since marginal contribution $\phi_k (k = 1, 2, \dots, 5)$ reflects the relative importance of DMs in group collaboration, its changes can reveal the influence of psychological preference differences on the group cooperation pattern and trust correction mechanism. By analyzing the dynamic changes in the marginal contributions of each DM under the variation of $\xi_k (k = 1, 2, \dots, 5)$, it can be verified whether the model can effectively depict the contribution differences caused by psychological preferences at the group level, as shown in Fig. 6.

As seen in Fig. 6, changes in psychological preference factors $\xi_k (k = 1, 2, \dots, 5)$ cause significant fluctuations in the Shapley ranking of different DMs. For instance, DM d_1 exhibits a higher contribution when $\xi_k (k = 1, 2, \dots, 5) = 0$ and $\xi_k (k = 1, 2, \dots, 5) \in [0.2, 0.4]$, but a lower contribution at other values, indicating that DM's influence fluctuates considerably across different psychological states. In contrast, DM d_5

Table 5
Trust relationship matrix $T = (t_{kh})_{5 \times 5}$ for DMs.

	d_1	d_2	d_3	d_4	d_5
d_1	$\times$	$ \psi_{12}\rangle = \sqrt{0.2} 0\rangle + \sqrt{0.6} 1\rangle$	$ \psi_{13}\rangle = \sqrt{0.5} 0\rangle + \sqrt{0.3} 1\rangle$	$ \psi_{14}\rangle = \sqrt{0.2} 0\rangle + \sqrt{0.3} 1\rangle$	$ \psi_{15}\rangle = \sqrt{0.6} 0\rangle + \sqrt{0.2} 1\rangle$
d_2	$ \psi_{21}\rangle = \sqrt{0.4} 0\rangle + \sqrt{0.4} 1\rangle$	$\times$	$ \psi_{23}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.4} 1\rangle$	$ \psi_{24}\rangle = \sqrt{0.7} 0\rangle + \sqrt{0.1} 1\rangle$	$ \psi_{25}\rangle = \sqrt{0.35} 0\rangle + \sqrt{0.45} 1\rangle$
d_3	$ \psi_{31}\rangle = \sqrt{0.25} 0\rangle + \sqrt{0.35} 1\rangle$	$ \psi_{32}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.2} 1\rangle$	$\times$	$ \psi_{34}\rangle = \sqrt{0.2} 0\rangle + \sqrt{0.8} 1\rangle$	$ \psi_{35}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.1} 1\rangle$
d_4	$ \psi_{41}\rangle = \sqrt{0.6} 0\rangle + \sqrt{0.2} 1\rangle$	$ \psi_{42}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.5} 1\rangle$	$ \psi_{43}\rangle = \sqrt{0.2} 0\rangle + \sqrt{0.2} 1\rangle$	$\times$	$ \psi_{45}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.4} 1\rangle$
d_5	$ \psi_{51}\rangle = \sqrt{0.5} 0\rangle + \sqrt{0.2} 1\rangle$	$ \psi_{52}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.4} 1\rangle$	$ \psi_{53}\rangle = \sqrt{0.3} 0\rangle + \sqrt{0.4} 1\rangle$	$ \psi_{54}\rangle = \sqrt{0.4} 0\rangle + \sqrt{0.3} 1\rangle$	$\times$

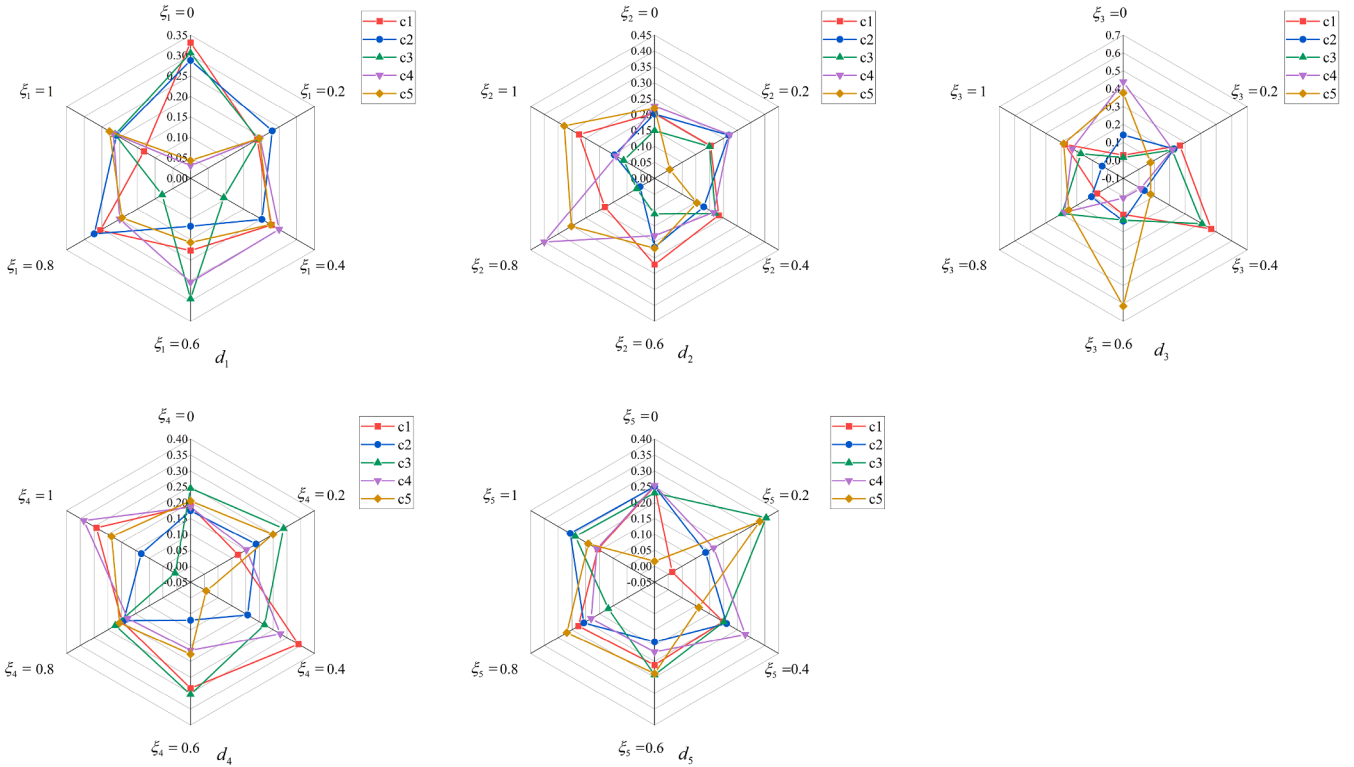


Fig. 5. Influence of psychological preference factors ξ_k ($k = 1, 2, \dots, 5$) on the weight of attributes.

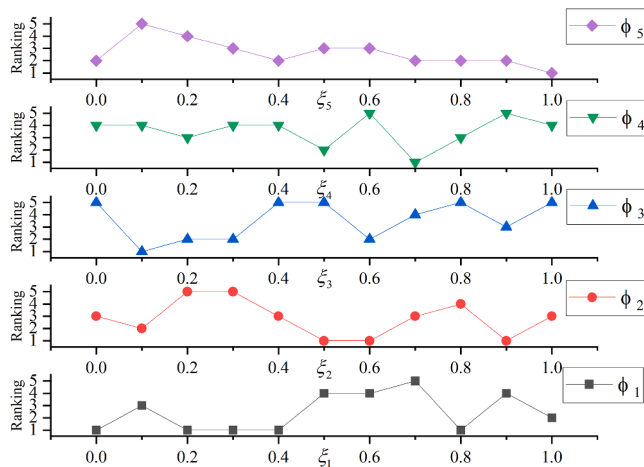


Fig. 6. Ranking changes of psychological preference factor ξ_k ($k = 1, 2, \dots, 5$) to describe contribution degree of DMs.

Table 6

The conditional probability of QLBN.

$a_i \backslash d_k$	d_1	d_2	d_3	d_4	d_5
a_1	0.2084	0.1481	0.1408	0.2517	0.1967
a_2	0.2598	0.1996	0.2564	0.2097	0.2061
a_3	0.1529	0.1668	0.0772	0.1069	0.0446
a_4	0.1291	0.3113	0.3577	0.1466	0.3032
a_5	0.2498	0.1741	0.1678	0.2850	0.2494

demonstrates high stability, consistently maintaining a strong contribution in each psychological state, suggesting that d_5 has a relatively strong and steady influence in group decision-making. The sensitivity analysis highlights the substantial impact of psychological preference factors on the contribution levels of DMs in group decision-making. It reveals that the marginal contribution of DMs shifts under varying psychological states.

3) The influence of trust modification model parameters on DMS' weights

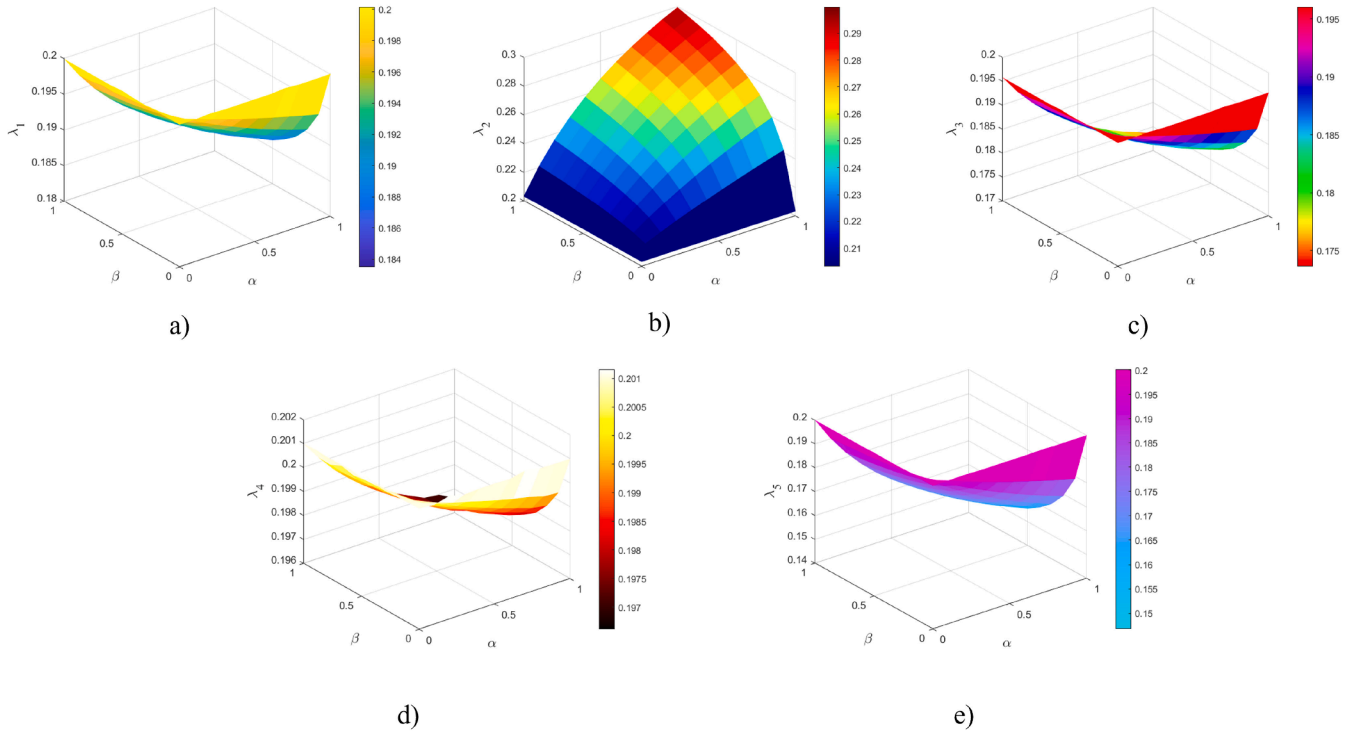


Fig. 7. Influence of regulatory factors α and weight coefficients β on the weight of DMs.

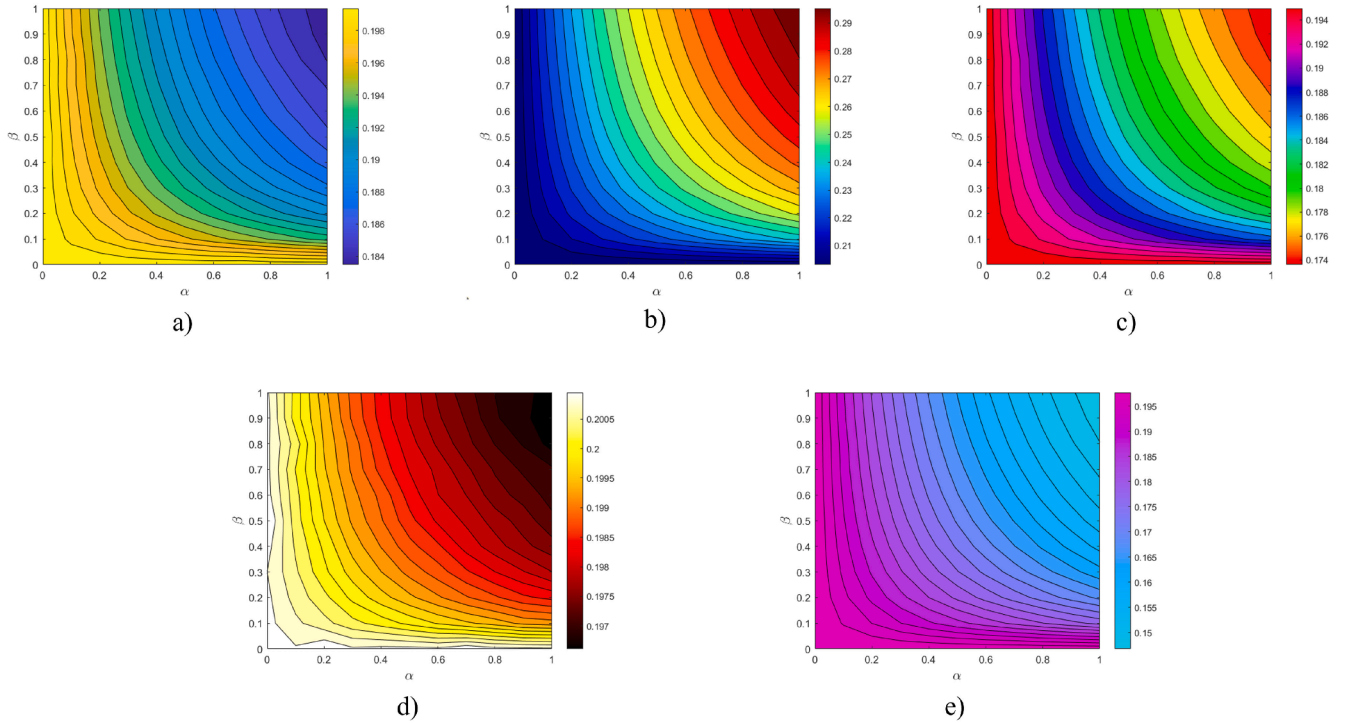


Fig. 8. Influence of regulatory factors α and weight coefficients β on the weight of DMs.

From the perspective of the trust correction mechanism, Model (7) describes the coupling relationship between the intensity of trust evolution and fair weighting through parameters (α, β) , where α controls the adjustment amplitude of trust's contribution, β reflects the retention effect of initial trust on modified trust, as shown in Fig. 7.

In order to observe and analyze the image more intuitively and clearly, the three-dimensional image is projected onto the two-dimensional plane, as shown in Fig. 8.

Fig. 8 illustrates the changing trend of each DM's weight λ_k with respect to the adjustment factor α and the weight coefficient β . The results indicate that the sensitivity of DMs to these parameters varies significantly. The weight of DM d_2 increases notably as α rises, suggesting that the weight of DM d_2 is more dependent on changes in the Shapley value. In contrast, DM d_4 shows greater sensitivity to changes in β , with its weight fluctuating due to the amplification effect of trust correction.

Table 7
The comparison of different models.

Method	Trust relationship representation	Psychological preference	Group cooperation	Interference effect	Angle measurement method
Zhu et al. (2025)	[0, 1]	×	×	×	×
Wu et al. (2023b)	0–1	×	×	×	×
Jiang et al. (2024b)	IFS	×	×	×	×
Liao et al. (2024)	[0, 1]	×	✓	×	×
She et al. (2021)	×	×	×	✓	Deng entropy
Wang et al. (2024)	[0, 1]	×	×	✓	Similarity heuristic method
Wu et al. (2022)	×	×	×	✓	Constant value
The proposed method	QIFS	✓	✓	✓	Kullback-Leibler Divergence

Moreover, some DMs exhibit localized sensitivity to the changes in the adjustment factor α and the weight coefficient β within specific regions, highlighting the complex and diverse responses of DMs to different parameter combinations during emergency situations. This sensitivity analysis provides a theoretical foundation for optimizing weight allocation strategies in emergency decision-making. It assists DMs in setting parameters more appropriately in the face of uncertainty and diverse information, ensuring the transparency, stability, and fairness of the decision-making process, and ultimately improving the overall efficiency and scientific rigor of complex meteorological disaster responses.

5.4. Comparative analysis

5.4.1. Qualitative comparative analysis of decision-making methods

This section examines the variances between the newly proposed model and earlier decision-making models from various angles, emphasizing the benefits of the suggested approach. Table 7 encapsulates the main differences between the proposed method and other comparative models.

Table 7 presents a comparative analysis between representative existing MAGDM models and the proposed method from four key dimensions: trust relationship representation, psychological preference modeling, group cooperation mechanism, and interference effect handling. The detailed comparison and improvements are as follows:

1) In terms of trust relationship representation, existing studies mostly adopt interval-based trust values (e.g., [0,1]) or traditional IFS to describe the trust level among DMs. However, these representations face limitations in capturing uncertainty, multi-state superposition, and the dynamic evolution of trust. In contrast, this paper proposes a QIFS, which integrates the concepts of quantum superposition and entanglement to naturally simulate the coexistence of multiple cognitive states and uncertain trust perceptions, thereby significantly enhancing the model's ability to represent dynamic trust relationships.

2) Regarding psychological preference modeling, most existing models overlook the dynamic influence of DMs' mental states on the perception of attribute weights or only reflect individual differences at the evaluation level, lacking systematic behavioral modeling. This paper introduces a Sigmoid-function-based psychological preference function, which incorporates psychological state as a regulating factor to dynamically adjust attribute weights. This mechanism captures the "neglect-emphasis-stabilization" evolution process and aligns with behavioral decision patterns such as marginal perception changes and risk sensitivity, thus improving the model's expressiveness regarding individual decision behavior.

3) In terms of group cooperation mechanism, while some studies address group preference aggregation, most fail to account for individual marginal contributions to the group objective. The proposed model incorporates a Shapley value-based cooperative game framework to evaluate each DM's contribution to group consensus, achieving a more equitable and reasonable weight distribution and promoting internal collaboration and consensus formation within the group.

4) In interference effect modeling and angle measurement, current models mainly adopt three types of approaches: (a) the fixed angle method, which assigns a constant phase angle to reflect interference strength—this approach is simple but fully subjective and lacks adaptability; (b) similarity-based heuristic methods, which estimate interference through similarity or distance measures between evaluations, yet lack theoretical grounding and are sensitive to data variations; and (c) entropy-based methods (e.g., Deng entropy), which quantify information uncertainty but cannot capture interference direction or differences in preference distributions. However, this paper uses the Kullback-Leibler (KL) divergence to measure the divergence between DMs' quantum probability distributions. This approach quantifies preference conflict from an information-theoretic perspective, with well-defined probabilistic semantics and directional sensitivity, making it more suitable for modeling mutual disturbances and interactive preferences among DMs in complex social networks.

5.4.2. Comparison of the ranking results of the alternatives

To assess the validity and robustness of the proposed model, a series of numerical experiments are carried out, in which the model is compared with several existing decision-making approaches. The resulting rankings derived from different methods are reported in Table 8.

As shown in Table 8, the optimal alternative obtained by Wan et al. (2022) is a_5 . Their method integrates group preferences through entropy weights and Jensen-Shannon divergence; however, the trust structure is static, and the weights cannot dynamically adjust with group interaction or psychological changes. In contrast, this paper achieves adaptive weight update and psychological feedback coupling based on quantum intuitionistic fuzzy numbers and a two-stage trust correction mechanism. Consequently, the resulting ranking better reflects the dynamic characteristics and stability of the group trust structure. The DEA regret cross-efficiency model proposed by Liu et al. (2024) also yields a_4 as the optimal alternative. Although it incorporates risk and regret mechanisms to handle utility differences, its efficiency matrix and risk parameters are fixed, lacking the ability to capture multi-agent trust fluctuations and cognitive interference effects. By introducing the Shapley marginal contribution and a two-stage trust correction model, this paper couples individual contributions with dynamic trust and quantifies interference effects within a quantum framework, thereby achieving coordinated optimization between trust feedback and behavioral consistency, demonstrating higher robustness and interpretability.

Compared with Quantum model 1 (Wu et al., 2022), although the sequencing result is entirely consistent with this model, it fails to accurately calculate the interference value among DMs, relying instead on a few representative values for calculation. This approach cannot fully capture the interference among DMs, making it less effective in handling complex and uncertain decision environments. Quantum model 2 (Jiang & Liu, 2022) identifies a_2 as the optimal alternative, but it does not fully consider the mutual trust relationships between DMs or the impact of their psychological preferences in complex decision situations. Moreover, this method uses cosine similarity to measure the interference value, which only accounts for vector directionality and overlooks actual score differences, leading to inaccurate measurements. The results

Table 8
Comparison of the results of different models.

Method	Ranking index	Ranking
SN-GDM 1 (Wan et al., 2022)	$h_1^* = 0.5037, h_2^* = 0.5357,$ $h_3^* = 0.3627, h_4^* = 0.6589,$ $h_5^* = 0.6905.$	$a_5 > a_4 > a_2 > a_1 > a_3$
SN-GDM 2 (Liu et al., 2024)	$U_1^* = 0.4799, U_2^* = 0.4835,$ $U_3^* = 0.4669, U_4^* = 0.4869,$ $U_5^* = 0.4839.$	$a_4 > a_5 > a_2 > a_1 > a_3$
Quantum model 1 (Wu et al., 2022)	$p(a_1) = 0.1899, p(a_2) = 0.2295,$ $p(a_3) = 0.1091, p(a_4) = 0.2455,$ $p(a_5) = 0.2260.$	$a_4 > a_2 > a_5 > a_1 > a_3$
Quantum model 2 (Jiang & Liu, 2022)	$p(a_1) = 0.1998, p(a_2) = 0.2391,$ $p(a_3) = 0.1165, p(a_4) = 0.2082,$ $p(a_5) = 0.2365.$	$a_2 > a_5 > a_4 > a_1 > a_3$
Quantum model 3 (She et al., 2021)	$p(a_1) = 0.3558, p(a_2) = 0.1719,$ $p(a_3) = 0.1533, p(a_4) = 0.1348,$ $p(a_5) = 0.1842.$	$a_1 > a_5 > a_2 > a_3 > a_4$
The proposed model	$p(a_1) = 0.1674, p(a_2) = 0.2177,$ $p(a_3) = 0.1325, p(a_4) = 0.2782,$ $p(a_5) = 0.2042.$	$a_4 > a_2 > a_5 > a_1 > a_3$

obtained from Quantum model 3 (She et al., 2021) show considerable differences when compared to the model introduced in this work. This approach employs Deng entropy to assess the interference value, emphasizing the variability of individual score distributions and neglecting the variations in score distributions among DMs, which results in skewed measurements. Additionally, the model fails to adequately account for the evolving trust dynamics among DMs within social networks.

However, this paper presents a two-phase trust adjustment scheme grounded in cooperative game theory, which integrates the contribution levels and trust degrees of DMs to dynamically modify their trust relationships within social networks. Furthermore, it takes into account the psychological preferences of DMs in complex and uncertain settings to ascertain attribute weights. Ultimately, the KL divergence is employed to quantify the interference value, adeptly reflecting the disparities among DMs and offering a more precise assessment of the interference angle, thereby enhancing its applicability in real-world decision-making contexts.

5.4.3. Ablation experiment on attribute weights based on psychological preference analysis

To verify the effectiveness and rationality of the proposed attributes' weights model considering psychological preferences in MAGDM, this section conducts a statistical simulation experiment to compare the performance differences between Model A and Model B. Model A corresponds to Model (7) in this paper, which explicitly incorporates a psychological preference function in the weight computation process, whereas Model B is derived from Model A by removing the psychological preference function while keeping all other structures and parameters identical. The purpose of this experiment is to examine whether, under identical structural conditions, the introduction of a psychological preference mechanism significantly alters the distribution characteristics of individual attribute weights, thereby influencing DMs' evaluations of attribute importance.

In the experiment, A decision matrix of 5×5 is randomly generated based on the Rand function. For each simulation experiment, this matrix is respectively substituted into Model A and Model B for solution to obtain the attribute weight vector of each DM:

$$\begin{aligned}\omega_k^A &= [\omega_k^{Ak}(\xi_k), \omega_2^{Ak}(\xi_k), \dots, \omega_5^{Ak}(\xi_k)], \\ \omega_k^B &= [\omega_1^{Bk}, \omega_2^{Bk}, \dots, \omega_5^{Bk}].\end{aligned}$$

To avoid randomness-induced bias, the entire process is independently repeated 100 times to generate 100 paired weight results. This paper adopts the following three indicators to measure the differences in the weight distribution of the two models:

Table 9
Wilcoxon signed-rank test results of the differences between Model A and Model B.

Metric	Median	Z	p-value	r	Significance
L_2	0.474	+8.68	$1.98E-18$	0.868	+
ΔH	-0.333	-8.60	$3.84E-18$	0.860	+

1) Euclidean Distance : This metric quantifies the overall magnitude of deviation between two weight vectors and reflects how strongly the introduction of psychological preferences alters the weight magnitudes.

It is given by: $L_2(\omega_k^A, \omega_k^B) = \left(\sum_{j=1}^n (\omega_j^{Ak}(\xi_k) - \omega_j^{Bk})^2 \right)^{1/2}$.

2) Entropy : This metric characterizes the impact of psychological preferences on the uniformity of weight distributions. It is defined as:

$$\Delta H_k = H(\omega_k^A) - H(\omega_k^B), \text{ where } H(\omega_k) = - \sum_{j=1}^n \omega_j^k \log \omega_j^k.$$

Based on these computations, statistical analysis is performed on the 100 simulation runs. For each run, the mean of the five DMs' results is calculated to obtain the global-level indicator sequence. Subsequently, the Wilcoxon signed-rank test (Rosner et al., 2005) is used to evaluate whether the median differences of these indicators are statistically significant.

The hypotheses of the test are formulated as: H_0 : There is no significant difference in the median weight distributions obtained from Models A and B, H_1 : The weight distributions obtained from Models A and B differ significantly.

For L_2 , a one-sided test H_1 : median > 0 is used, while for ΔH , H_1 : median < 0 is adopted. The significance level is set to $\alpha = 0.05$, and the effect size is calculated as $r = |Z|/\sqrt{n}$ to measure the strength of the effect. The statistical results of the three metrics are presented in Table 9.

As can be seen from Table 9, the p values of each indicator are all much less than 0.05, and the effect size r is greater than 0.8, indicating that the influence of psychological preference on the weight distribution is extremely significant in statistical sense and has a very strong effect intensity. Specifically, the significant positive value of L_2 indicates that after the introduction of psychological preferences, both the form and numerical amplitude of attribute weights have undergone significant shifts, and the ranking of the importance of different attributes by DMs has been significantly changed. The significant negative value of ΔH indicates that the psychological preference mechanism reduces the entropy of the weight distribution, enabling DMs to focus more on key attributes. Overall, the research results confirm that the psychological preference mechanism plays a significant role in the for-

Table 10
Structural comparison of representative QLBN.

Reference	Decision environment	Prior probability	Conditional probability	Interference measure
Wang et al. (2024)	Crisp	Empirical trust values	Normalized trust links	Similarit heuristic method
Wu et al. (2022)	Fuzzy	Known	Evaluation information	Known
She et al. (2021)	Crisp	Evaluation information	Evaluation information	Deng entropy
This paper	Fuzzy + Dynamic	Trust revision mechanism	Psychologically adjusted attributes' weights and evaluations	Symmetric KL divergence

mation of attribute weights and significantly verify the model's ability to represent subjective preference features in the real decision-making process.

5.4.4. Comparative analysis of QLBN

To further visually and systematically demonstrate the differences between the quantum-like Bayesian network constructed in this paper and the existing representative QLBN models, we have supplemented a comparative analysis in the revised draft and made comparisons from key structural elements such as the decision-making environment, the source of prior probability, the construction method of conditional probability, and the measurement method of interference angle or interference intensity, as shown in Table 10. These dimensions all directly correspond to the core components of QLBN and are helpful for objectively evaluating the differences in modeling assumptions and structural designs among various models.

As summarized in Table 10, the proposed QLBN differs from existing QLBN frameworks in several key structural aspects, including the decision environment setting, the construction of prior and conditional probabilities, and the measurement of interference intensity. These differences are detailed as follows:

1) From the perspective of decision environment settings, existing QLBN models are mostly developed for crisp decision environments or scenarios with known and static fuzzy information, where evaluation data are typically represented by crisp values or predefined fuzzy numbers. In contrast, the proposed approach operates in a fuzzy and dynamic decision environment. Although the evaluation information in this paper is expressed in crisp form, fuzziness and uncertainty are explicitly incorporated at the level of trust relationships. Specifically, quantum intuitionistic fuzzy numbers are employed to represent interpersonal trust among DMs, allowing trust, distrust, and hesitation to coexist within a unified state representation. Moreover, these trust relationships are not static inputs but are dynamically updated through a two-stage trust correction mechanism, which enables the decision environment to evolve endogenously with group interactions.

2) Regarding the determination of prior probabilities, most existing models treat the prior probabilities of DM nodes as known quantities or derive them through direct calculation, normalization. Consequently, the resulting prior probabilities are only weakly connected to the evolution of relational structures among DMs. In contrast, the proposed QLBN constructs prior probabilities based on a trust network and its revision mechanism, allowing DM weights to be dynamically adjusted in response to changes in trust relationships. This design endows the prior probabilities with explicit behavioral meaning and update capability.

3) With respect to the construction of conditional probabilities, existing QLBN studies typically derive conditional probabilities directly from crisp or known evaluation information via simple aggregation or normalization, with limited consideration of psychological states or attribute weight variations. In the proposed framework, conditional probabilities are constructed within a multi-attribute evaluation setting, where attribute weights are dynamically adjusted by a psychological preference function. As a result, the conditional probabilities can reflect comprehensive evaluation differences across alternatives under varying psychological states of DMs.

4) Concerning the measurement of interference angles or interference intensity, existing studies predominantly rely on heuristic similarity measures, entropy-based indicators, or preset parameters. Such

approaches mainly describe similarity or uncertainty among evaluation outcomes, but they do not directly characterize the overall divergence structure between preference distributions, leading to an indirect link to distribution-level differences. In contrast, this study employs symmetric KL divergence to quantify the informational deviation between DMs' preference distributions and further maps it into an interference angle. This procedure allows interference intensity to be explicitly derived from distributional differences, thereby establishing a clearer quantitative connection between preference distribution structures and quantum interference terms.

Overall, existing QLBN models primarily emphasize the representation of interference forms or interference strength itself, whereas the proposed QLBN focuses on the internal generation mechanism and consistency of probabilities. By introducing fuzzy trust relationships on the basis of crisp evaluation information and systematically integrating trust, psychological preference, and multi-attribute evaluation mechanisms into the QLBN, the proposed framework not only captures interference effects among DMs but also reflects how such effects are formed and evolve during the group decision-making process.

6. Discussion

This paper focuses on the severe forest fire that broke out in April 2024 in Kunming City, Yunnan Province—a real emergency decision-making case in which extreme meteorological conditions caused rapid fire spread and posed multiple threats to both the environment and public safety. The decision process involved meteorological agencies, emergency management authorities, firefighting units, ecological protection offices and local government, and faced challenges such as tight time constraints, information asymmetry, limited resources and conflicting objectives. Therefore, we established a multi-mechanism and multi-stage MAGDM model based on this real scenario to provide scientific, precise and efficient emergency service decision support for public institutions, thereby enhancing the rapid response capability and social risk control.

6.1. The problem of group decision-making in emergency rescue of meteorological disasters

MAGDM in the context of meteorological disasters is a high-pressure, dynamic, and multi-agent process involving diverse and heterogeneous participants (https://www.gov.cn/lianbo/bumen/202501/content_6997126.htm; https://www.gov.cn/gongbao/content/2019/content_5468897.htm). It is characterized by sudden onset, urgency, and strong behavioral coupling. In such scenarios, group members must collaboratively evaluate multi-attribute alternatives under tight time constraints while facing cognitive divergence, behavioral interference, and evolving trust dynamics. The outcome of such decisions directly affects the efficiency of emergency response and the level of social risk control.

Specifically, the complexity of this type of decision-making is reflected in the following four aspects: 1) Strong dynamic game behavior. Participants are often from different functional departments or organizations (such as meteorological agencies, emergency command centers, and medical units), with differing goals and resource priorities. 2) Non-linear psychological preference fluctuations. Under the high-risk conditions of a disaster, DMs' psychological states and risk perceptions

change non-linearly with evolving information, time pressure, and contextual cues. 3) Evolving trust relationships. As collaboration deepens and information exchange continues, interpersonal trust levels adjust accordingly. 4) Significant mutual interference effects. Individual judgments in group settings are rarely made in isolation; they are influenced by others' opinions, perceived consensus paths, and psychological suggestibility.

6.2. Policy recommendations and management strategies

Based on the research on trust evolution, psychological preference fluctuations, and multi-party collaboration in meteorological disaster emergency group decision-making (https://www.ndrc.gov.cn/fzggw/jgsj/yxj/sjdt/202112/t20211231_1311303.html; <http://www.sasac.gov.cn/n2588025/n2643314/c34048720/content.html>), this paper proposes three practical policy recommendations:

1) Establish a dynamic trust evaluation and adjustment mechanism. On the emergency decision-making platform, based on the regular quantification of the internal trust level, the historical collaboration records and current decision making performance of each stakeholder. Use these trust metrics to flexibly adjust participants' decision-making roles and resource allocations, preventing collaboration breakdowns caused by trust imbalances or information gaps and thus improving overall decision efficiency and coordination.

2) Implement a dynamic preference and interference identification-response mechanism. Recognizing that DMs' risk preferences and judgments fluctuate with new information and group interactions, sometimes leading to convergence or conflict of opinions, so real-time monitoring and trend analysis of the attribute assessment of each participant are suggested to introduce. When significant preference shifts or group-level interference signals are detected, adjust the agenda (e.g., through phased discussions or subgroup evaluations) to ensure diverse viewpoints are fully heard, thereby safeguarding the stability and fairness of the decision process.

3) Formulate a cooperative incentive and coordination system towards marginal contribution. In emergency situations with multiple institutions and limited resources, it is recommended to design a cooperative incentive program towards marginal contribution. Resources and rewards are allocated based on the actual contributions of all parties in the implementation stages. Meanwhile, establish cross-departmental coordination agreements and clarify responsibilities, formulate resource sharing agreements and reward and punishment rules, institutionalized collaboration, etc, which will achieve the best resource coordination and maximum collective benefits.

6.3. Promotion and applicability in other fields

The proposed method in this paper can be effectively extended to other domains that involve multi-party collaboration, behavioral interaction, and uncertainty. This scalability arises from the model's integrated structure combining psychological preference adjustment, trust evolution, and cooperative contribution mechanisms, which together provide a dynamic and interpretable framework for analyzing decision behaviors under complex and interdependent conditions. Such features enable the proposed framework to be applied in various decision contexts, for example, in medical emergency response and supply-chain collaboration.

1) In medical emergency decision-making, hospitals, health authorities, and rescue teams often face severe time constraints and incomplete information when allocating limited resources. The proposed framework can describe the nonlinear evolution of medical experts' risk perceptions and mutual trust through the quantum intuitionistic fuzzy representation, while the Shapley-based contribution mechanism ensures fairness and transparency in cross-institutional coordination. This facilitates efficient and balanced collaboration in high-stakes medical rescue scenarios.

2) In supply-chain collaboration, multiple enterprises face uncertainty in demand, information asymmetry, and varying cooperation incentives. By modeling partners' trust evolution and psychological preferences through the quantum framework, and by applying Shapley-based contribution assessment, the proposed method supports stable and fair cooperation among supply-chain nodes.

7. Conclusion

In this paper, a fuzzy group decision-making method integrating cooperative game theory and quantum cognition is proposed to address the psychological differences, trust fluctuations, and interference effects in MAGDM within social networks. The method constructs a comprehensive framework from three perspectives. Firstly, a psychological preference perception function is developed to couple the psychological state of DMs with the attribute weighting process, dynamically capturing the nonlinear impact of psychological factors on weight allocation. Secondly, a QIFS is defined to characterize the fuzziness and dynamics of trust relationships, and a two-stage trust correction model is designed to describe the interactive evolution mechanism between trust and contribution. Finally, the KL divergence is used to measure interference effects among DMs, and the quantum probabilities of alternatives are calculated and ranked within a QLBN.

The research results show that: 1) the proposed method effectively captures the dynamic coupling relationship between psychological preference and trust evolution, whose interaction significantly influences the distribution of group weights. 2) the quantum interference mechanism plays an important role in modeling interdependent cognition among DMs, revealing how psychological variations affect trust adjustment and ranking stability. 3) The QIFS trust relationship representation and the two-stage trust correction method can describe the fuzziness and time-variability of trust relationships, making up for the deficiencies of traditional models in dynamic trust modeling.

However, the proposed method also has some limitations. First, in practical MAGDM problems, there may be complex interactions between the psychological preferences of experts and trust relationships. The impact of these interactions on decision outcomes requires further in-depth study. Additionally, the method's computational complexity increases when dealing with high-dimensional data and large-scale decision environments, potentially affecting practical application efficiency. On this basis, future research will be further expanded from the following three aspects:

1) Explore the dynamic linkage mechanism between psychological preference functions and the evolution process of trust. By introducing multi-layer feedback structures or collaborative learning strategies based on game balance, reveal the nonlinear influence paths among psychological fluctuations, trust correction and the formation of group consensus.

2) It is planned to introduce granularity constraints and uncertainty conservation mechanisms in quantum fuzzy representation to maintain the consistency of trust fuzziness and quantum interference during the evolution process, which is conducive to maintaining the robustness and information invariance of the trust evolution process in a multi-granularity and dynamically uncertain environment.

3) In response to the computational pressure of the current model in high-dimensional and large-scale scenarios, in the future, sparse matrix factorization, distributed parallel computing and approximate reasoning algorithms will be combined to reduce the time complexity and storage cost of the model, thereby enhancing its real-time performance and scalability.

CRedit authorship contribution statement

Shuli Yan: Writing – original draft, Methodology, Conceptualization; **Yizhao Xu:** Writing – original draft, Writing - review & editing,

Visualization, Methodology; **Jian Liu**: Writing – review & editing, Supervision; **Francisco Javier Cabrerizo**: Conceptualization, Writing – review & editing, Supervision.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proofs

A.1. Proof of theorem 1

Proof. Consider the function $f(\xi_k) = \frac{1}{1+e^{-\alpha(\xi_k-\xi_0)}}$. Obviously, $f(\xi_k)$ is monotonically increasing, because $\frac{d}{d\xi_k} f(\xi_k) = \frac{\chi e^{-\chi(\xi_k-\xi_0)}}{(1+e^{-\chi(\xi_k-\xi_0)})^2} > 0$. So $f(\xi_k)$ increases as ξ_k increases. By $\varpi_j^k(\xi_k) = \varpi_j^k \cdot f(\xi_k)$, we know $\varpi_j^k(\xi_k) = \frac{\varpi_j^k}{1+e^{-\chi(\xi_k-\xi_0)}}$. Since $f(\xi_k)$ is monotonically increasing, its reciprocal $\frac{1}{f(\xi_k)} = 1+e^{-\chi(\xi_k-\xi_0)}$ is monotonically decreasing. Further, considering that $\varpi_j^k(\xi_k)$ is a non-negative linear transformation of $f(\xi_k)$, $\varpi_j^k(\xi_k)$ is also monotonically decreasing, that is, $\xi_k > \xi'_k \Rightarrow \varpi_j^k(\xi_k) \leq \varpi_j^k(\xi'_k)$, so the theorem is proved.

A.2. Proof of corollary 1

Proof. Firstly, consider that the second derivative of the mental preference function is $\frac{d^2}{d\xi_k^2} f(\xi_k) = \frac{\chi^2 e^{-\chi(\xi_k-\xi_0)}(e^{-\chi(\xi_k-\xi_0)}-1)}{(1+e^{-\chi(\xi_k-\xi_0)})^3}$, since for all $\xi_k(k=1,2,\dots,s)$, the second derivative of this function is positive, that is $\frac{d^2}{d\xi_k^2} f(\xi_k) > 0$, therefore $f(\xi_k)$ is convex function. Since $\varpi_j^k(\xi_k) = \varpi_j^k \cdot f(\xi_k)$ and $\varpi_j^k > 0$, $\varpi_j^k(\xi_k)$ is also convex function.

A.3. Proof of theorem 2

Proof. Firstly, consider that the decision variable in the model is the attribute weight $\varpi_j^k(\xi_k)$ for each DM. According to the constraints of the model, all weights must satisfy the normalization conditions $\sum_{j=1}^n \varpi_j^k(\xi_k) = 1$, and $\varpi_j^k(\xi_k) \geq 0$. These constraints indicate that the set of feasible solutions for the weights lies within a standard simplex, which is a closed and bounded set in a higher-dimensional space. Therefore, the feasible solution space of the model forms a compact, closed set that is finite and complete.

Secondly, the objective function of the model is a weighted summation function based on the Shapley value. The Shapley value is essentially a weighted average of the marginal contributions, while the utility function $V(S)$ in the model is a nonlinear function of the decision variable $\varpi_j^k(\xi_k)$. Since both the foreground value v_{ij}^k and the decision weight function $\pi_{ij}^k(\varpi_j^k(\xi_k))$ are continuous and differentiable over the feasible domain of the decision variable, the objective function, as a combination of these continuous functions, is continuous on a compact set.

According to the Weierstrass theorem (Cotrino & Flores-Bazán, 2024), any continuous function defined on a compact and non-empty domain must have a global optimal solution. Since the feasible solution space of the model is compact and the objective function is continuous within this space, it follows that the model possesses an optimal solution.

A.4. Proof of theorem 3

Proof. Since the quantum states $|\psi_{x1}\rangle$ and $|\psi_{x2}\rangle$ are unit vectors, the modular square $|\langle\psi_{x1}|\psi_{x2}\rangle|^2$ of the inner product has a value between $[0,1]$. When the two quantum states are exactly the same, the modular square of the inner product is 1, indicating the maximum similarity, and the distance is 0. When the two quantum states are completely orthogonal, the modular square of the inner product is 0, indicating no similarity, and the distance is 1. Since $|\langle\psi_{x1}|\psi_{x2}\rangle|^2$ is in $[0,1]$, $1 - |\langle\psi_{x1}|\psi_{x2}\rangle|^2$ is also in $[0,1]$, and further through the square root function, the value of the distance formula $d(Q_1(x), Q_2(x))$ is always non-negative and at maximum, so the range of distances is strictly restricted to between $[0,1]$.

Appendix B. Background description of DMs

The decision-making team in this paper consists of five DMs, whose backgrounds and decision-making responsibilities are as follows:

1) d_1 : From the Emergency Management Department of Yunnan Province, has long been engaged in the formulation of emergency plans, command and dispatch, and the construction of emergency response systems, and has rich experience in handling major natural disasters. In decision-making, it mainly focuses on the coordination efficiency of the command chain of the plan, the rationality of resource scheduling and the feasibility of emergency response, representing the balancing position of government departments in comprehensive coordination and risk control.

2) d_2 : From a forestry and ecological research institution, with many years of experience in forest ecosystem restoration and fire risk assessment research, and has participated in multiple forest fire prevention and ecological protection projects. Its main concerns in the model are the ecological consequences of fire scene intervention, the vegetation recovery cycle and sustainable management strategies.

3) d_3 : From the Comprehensive Emergency Rescue Detachment of Yunnan Province, with rich experience in forest fire fighting and on-site command of emergency rescue, and has participated in cross-regional fire response and emergency drills many times. In the decision-making process, particular attention should be paid to the on-site operability of the rescue plan, safety protection measures and personnel risk control, representing the practical perspective of the front-line execution department.

4) d_4 : From the Yunnan Meteorological Bureau, specializing in fire meteorology, wind field dynamic simulation and geographic information System (GIS) integrated analysis, and has led multiple provincial and ministerial-level meteorological disaster assessment projects. It is responsible for evaluating the fire spread prediction accuracy, meteorological adaptability and spatial scheduling rationality of the scheme in group decision-making, providing scientific prediction support for the model.

5) d_5 : From the Kunming Social Work Federation, with rich experience in grassroots emergency management and mobilization of social

organizations, familiar with post-disaster material coordination and resident resettlement mechanisms. In group decision-making, the main focus is on the fairness of social rescue, the acceptability of residents' resettlement, and the transmission mechanism of public trust, representing the interests and demands at the social and public levels.

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